



Data Engineering Patterns

AWS Academy Cloud
Architecting



Introduction

Data Engineering Patterns

Module objectives

- This module prepares you to do the following:
 - Use the AWS Well-Architected Framework to generalize the type of architecture that is required to suit common use cases for data ingestion (batch and stream).
 - Select a data ingestion pattern appropriate to characteristics of the data (velocity, volume, and variety).
 - Select the appropriate AWS services to ingest and store data for a given use case.
 - Select the appropriate AWS services to optimize data processing and transformation requirements for a given use case.
 - Identify when to use different types of AWS data analytics and visualization services based on a given use case.

Module overview

Presentation sections

- Data characteristics
- Data pipelines
- AWS tools to ingest data
- Processing batch data
- Processing real-time data
- Storage in the data pipeline
- Parallel processing in the data pipeline
- Analysis and visualization
- Applying the AWS Well-Architected Framework principles to data pipelines

Activities

- Choosing Data Storage for a Bank Application
- Data Pipeline Architecture

Knowledge checks

- 10-question knowledge check
- Sample exam question

As a cloud architect designing a data architecture:



- I need to evaluate the characteristics of available data so that I can choose components that suit the data and the desired business outcomes.
- I need to think about how to store, organize, and access the data to get the most value for the business.

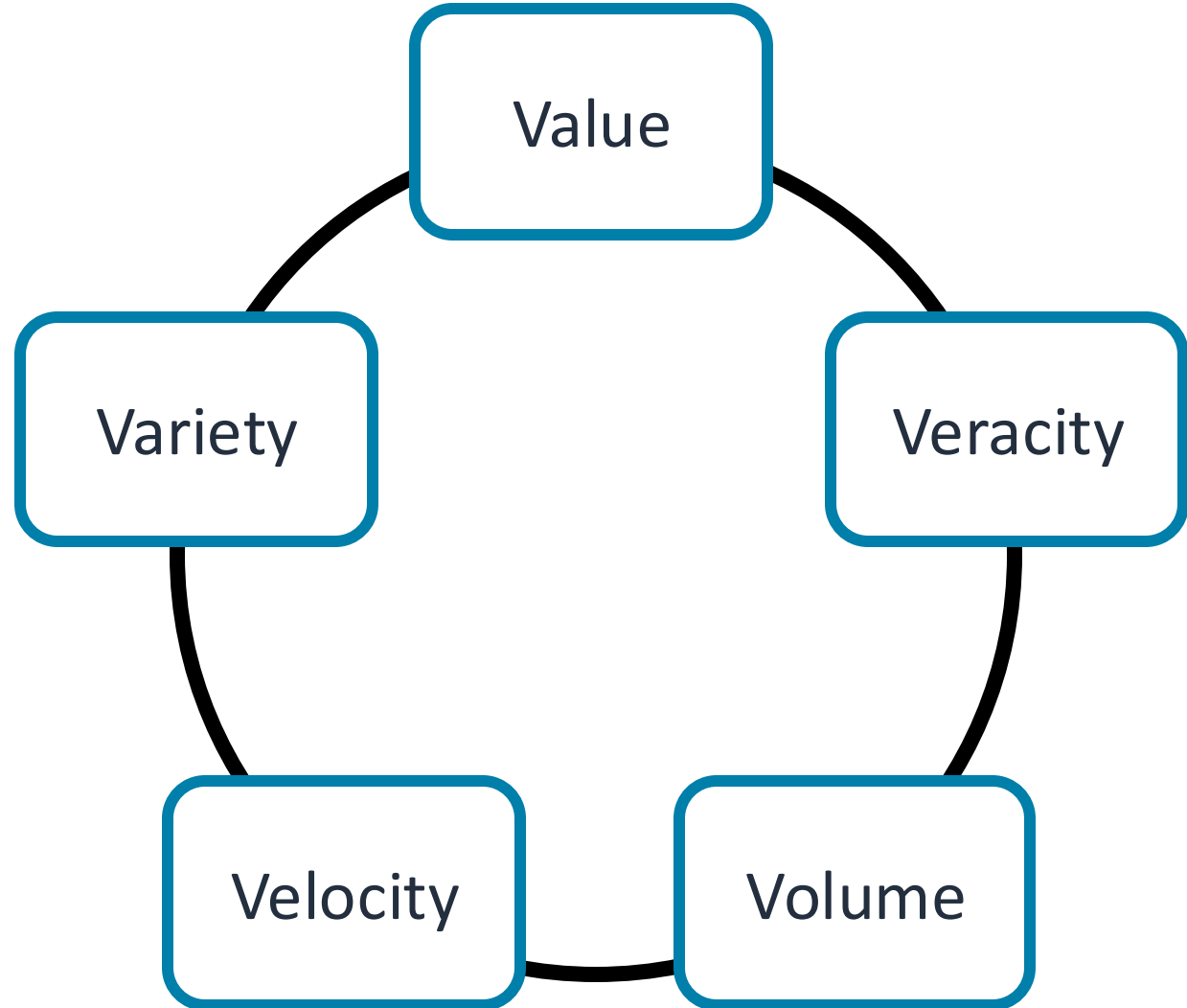


Data characteristics

Data Engineering Patterns

Data characteristics that drive infrastructure decisions

Consider the data characteristics together to make decisions for each business use case that you design a data infrastructure for.



Five Vs of data

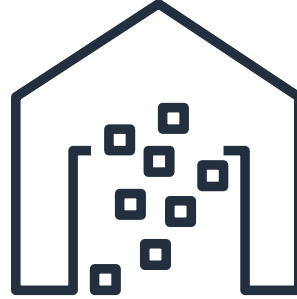
Data Characteristics	Scope	Questions to Consider
Value	How processed data can provide insight into a business problem	What insights can be gained from the data?
Veracity	How to protect and strengthen the integrity of data	How accurate, precise, and trusted is the data?
Volume	How much data you need to process	How long do you need to keep the data? What are the access patterns?
Velocity	How quickly data enters and moves through your pipeline	How frequently is data generated? How quickly does the data need to be acted on?
Variety	How many data sources and data types you work with	What is the format and type of the data? What sources does the data come from?

Three-pronged strategy to build data infrastructures



Modernize

Move to a cloud-based infrastructure and purpose-built services to reduce undifferentiated lifting.



Unify

Create a single source of truth for data, and make the data available across the organization.



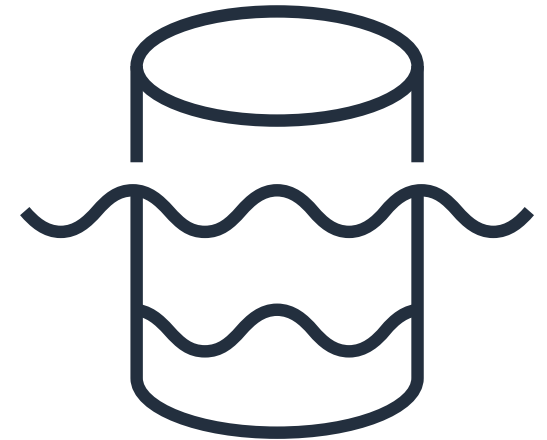
Innovate

Apply artificial intelligence and machine learning (AI/ML) to find new insights in the data.

A solution for a modern data architecture

Centralized location to access data and run analytics and AI/ML applications.

- Integrates a data lake, a data warehouse, and other purpose-built data stores
- Supports unified access, permissions, authorization, and seamless data movement
- Provides access to all of the data to make better decisions with agility



Data lake

Key takeaways: Data characteristics



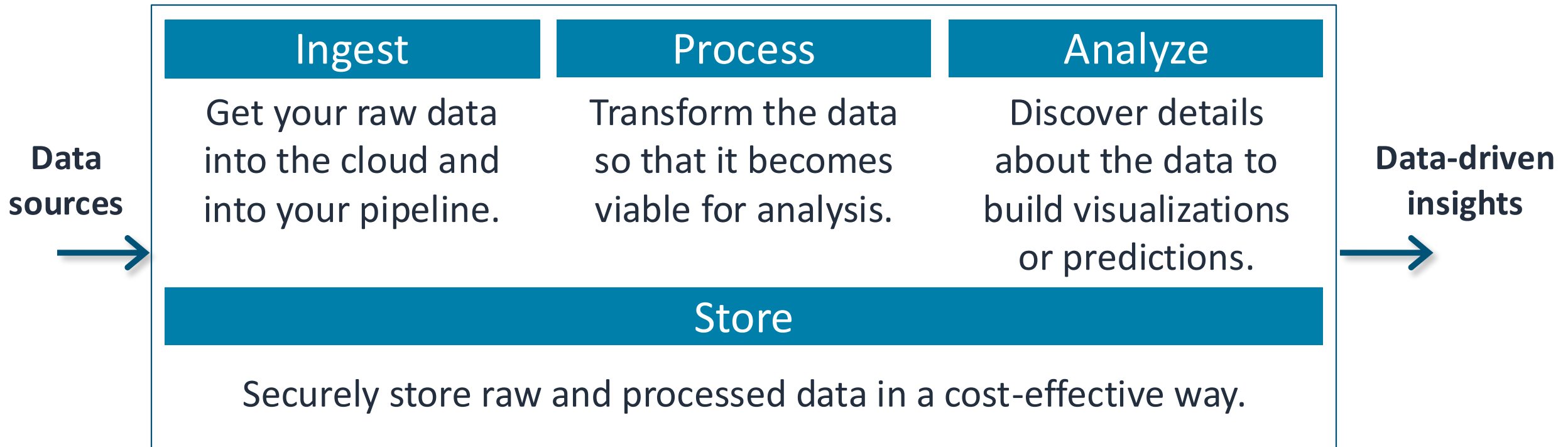
- Five data characteristics are value, veracity, volume, velocity, and variety.
- Consider the data characteristics together to make decisions about the infrastructure design for each business use case.
- Value and veracity impact accurate insights that the business needs. Volume and velocity *together* drive the expected throughput and scaling requirements of your pipeline.
- The goal of the modern data architecture is to store data in a centralized location and make it available to all consumers to perform analytics and run AI/ML applications.



Data pipelines

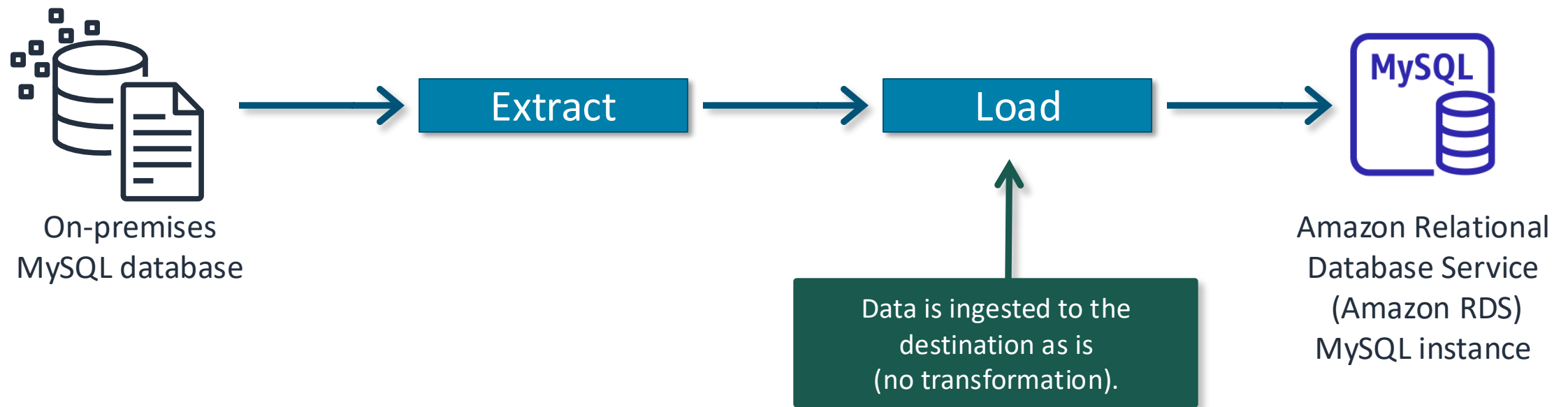
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Elements of a data pipeline



Homogeneous ingestion pattern

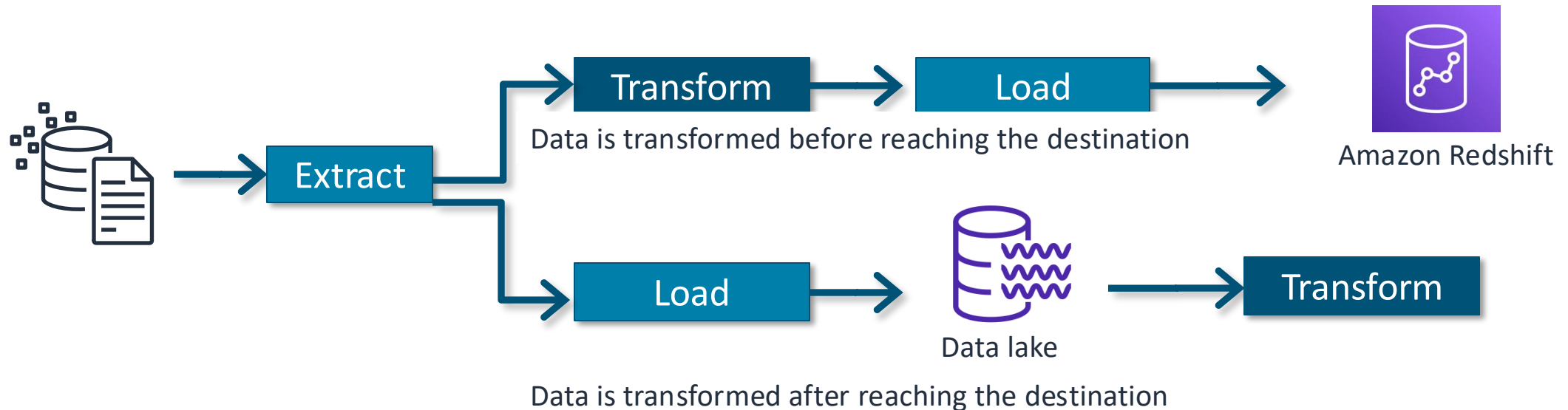
The primary objective of homogenous ingestion is to move data from a source data store to a destination data store, while keeping the same data format or data storage engine type.



Heterogeneous ingestion patterns

Extract, transform, and load (ETL)

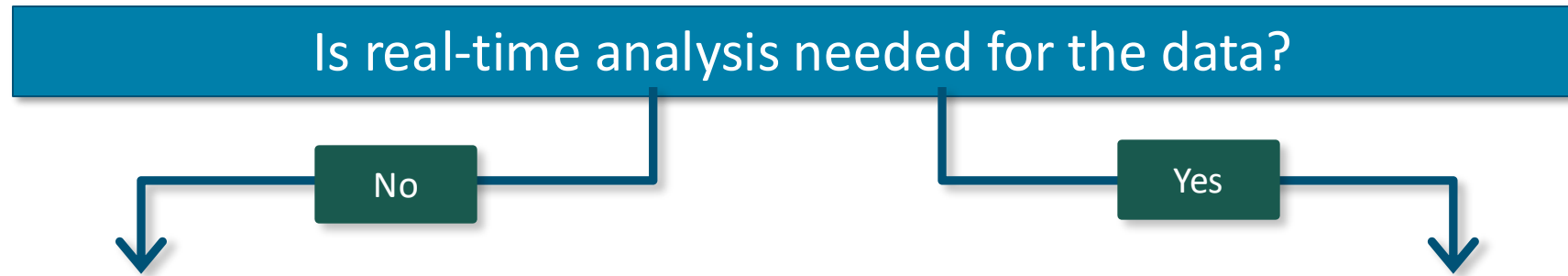
- Works well with structured data that is destined for a data warehouse
- Stores data that is ready to be analyzed, so this pattern can save time for an analyst



Extract, load, and transform (ELT)

- Works well for unstructured data that is destined for a data lake
- Offers flexibility to create new queries (analysts can access more raw data)

Batch and streaming processing patterns



Batch ingestion and processing

- Batch processing usually computes results based on complete datasets.
- Batch pipelines support deep analysis of large datasets.

Streaming ingestion and processing

- A data stream is unbounded; it's a continuous, incremental sequence of small data packets.
- Streaming pipelines process a series of events for real-time analytics.

Comparison of batch and streaming processing patterns

Features	Batch Processing	Streaming Processing
Data Processing Cycles	Processing pipelines run infrequently and typically during off-peak hours	Processing pipelines run continuously
Compute Requirements	Requires high computing power	Requires low computing power and reliable, low-latency network connections
Use Case	Sales transaction data is analyzed overnight, and reports are sent to branches in the morning.	When providing a product recommendation, data must be analyzed immediately.

Key takeaways: Data pipelines



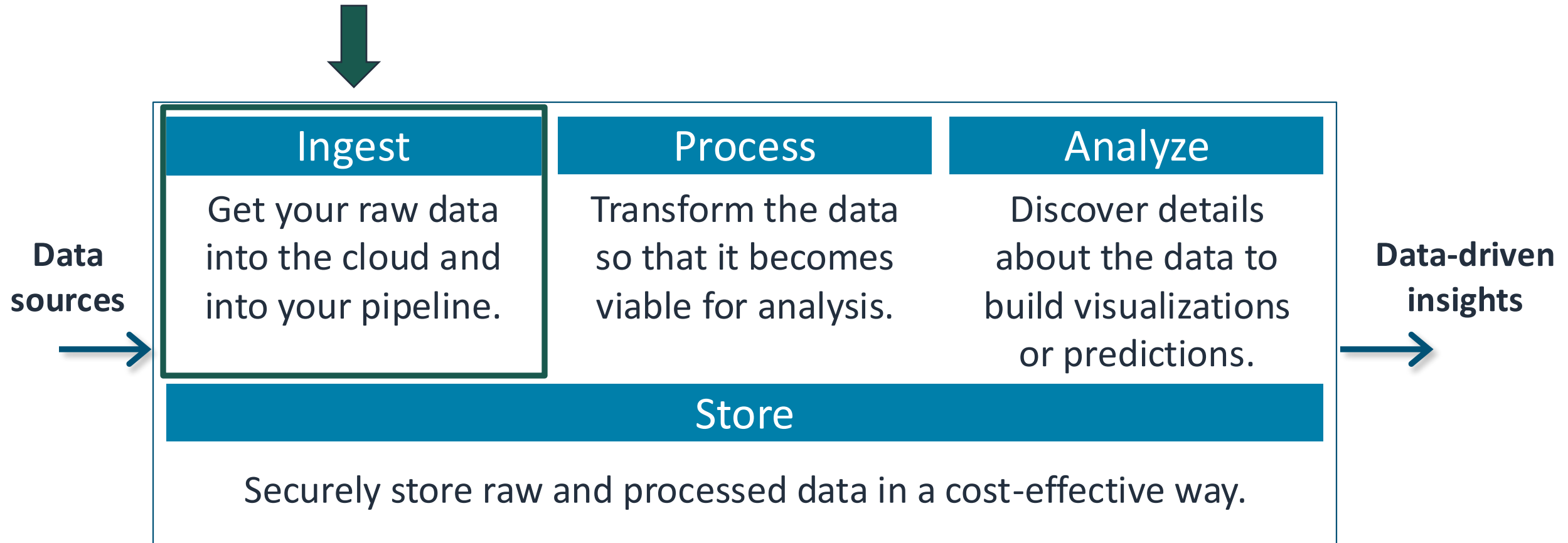
- A data pipeline includes layers to ingest, store, process, and analyze data.
- Data ingestion is the process of extracting data from its source and loading it into a data pipeline to be stored or analyzed.
- In homogenous ingestion, data doesn't need to be transformed because the source and destination match. When transformation is needed, heterogenous ingestion can involve ETL or ELT patterns.
- If the data needs to be analyzed in near real time, consider streaming ingestion methods instead of batch ingestion.



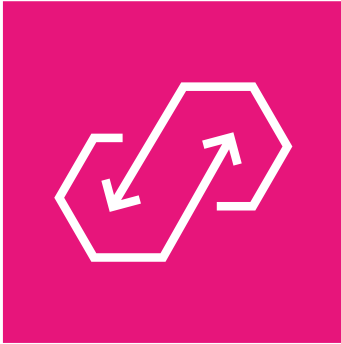
AWS tools to ingest data

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Ingestion layer of the data pipeline



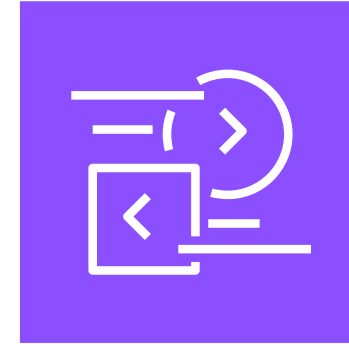
Matching ingestion services to data sources



Amazon AppFlow



AWS DataSync



AWS Data Exchange



Software as a service (SaaS)
applications

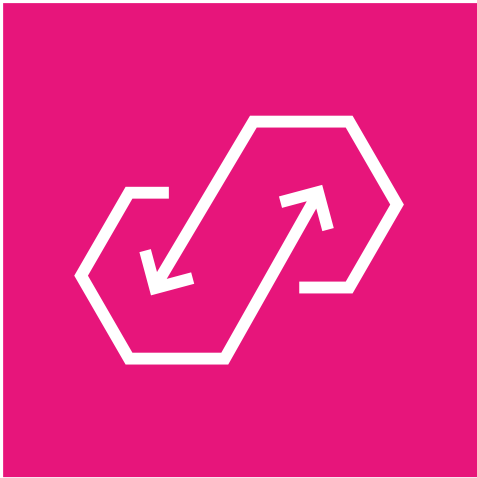


File shares



Third-party data

Amazon AppFlow



Amazon AppFlow

- Provides the ability to transfer data between SaaS applications and AWS services
- Offers reuse of available service integrations with available Amazon AppFlow APIs
- Provides data transformation capabilities

AWS DataSync



DataSync

- Is a fully managed data migration service
- Simplifies, automates, and accelerates copying file and object data to and from AWS storage services
- Is optimized for speed
- Includes encryption and integrity validation
- Preserves metadata when moving data

AWS Data Exchange



AWS Data Exchange

- Provides customers with a way to find, subscribe to, and use third-party data in the cloud
- Bridges the gap between providers and subscribers who exchange data by supporting data delivery through files, tables, and APIs
- Simplifies finding, preparing, and using data in the cloud

Key takeaways: AWS tools to ingest data



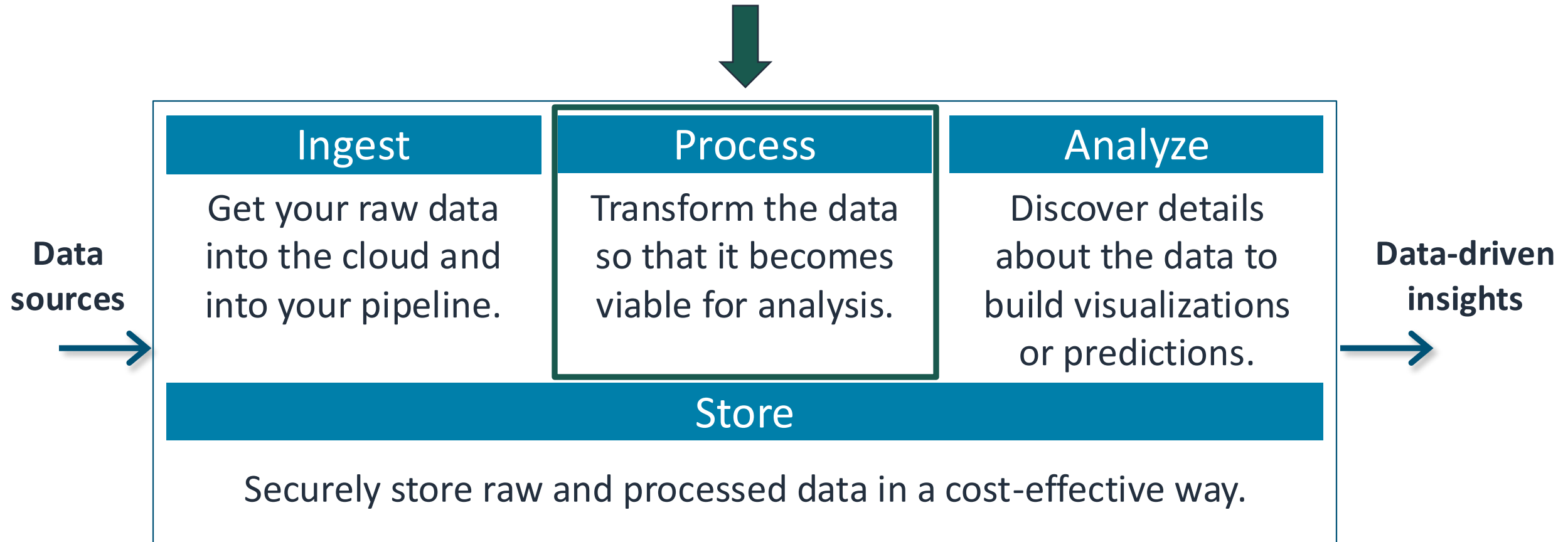
- To ingest data, consider purpose-built tools that will reduce undifferentiated lifting.
- Amazon AppFlow can ingest from SaaS applications, such as Salesforce and Zendesk.
- DataSync can ingest from file shares.
- AWS Data Exchange provides customers with a way to find, subscribe to, and use third-party data in the cloud.



Processing batch data

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Processing layer of the data pipeline



Batch ingestion and processing

Consider a banking system that receives financial transactions throughout the day.

The system collects and processes all the financial transactions at the end of each day to provide reports to stakeholders and third-party systems.



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graph TD; A[Periodically complete high-volume, repetitive data jobs] --> B[Consider a banking system that receives financial transactions throughout the day. The system collects and processes all the financial transactions at the end of each day to provide reports to stakeholders and third-party systems.]; C[Used for daily and weekly reporting] --> B; B --> D[Compute-intensive tasks are processed in batches, often in off-peak times]
```

Periodically complete high-volume, repetitive data jobs

Used for daily and weekly reporting

Compute-intensive tasks are processed in batches, often in off-peak times

When to consider batch processing

- For reporting purposes
- When dealing with large datasets
- When the analytics use case is focused more on aggregating or transforming data and less on real-time analysis of data



AWS Glue



AWS Glue

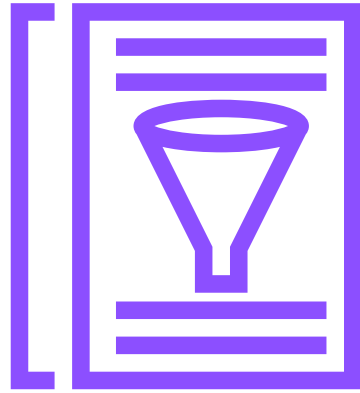
- Is a data integration service that helps automate and perform ETL tasks as part of ingesting data into a pipeline
- Provides the ability to read and write data from multiple systems and databases
- Simplifies batch and streaming ingestion

AWS Glue components



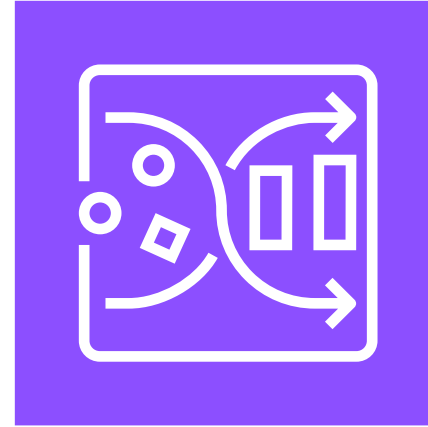
ETL jobs

- Data transformation jobs



AWS Glue Data Catalog

- Data discovery and organization



AWS Glue DataBrew

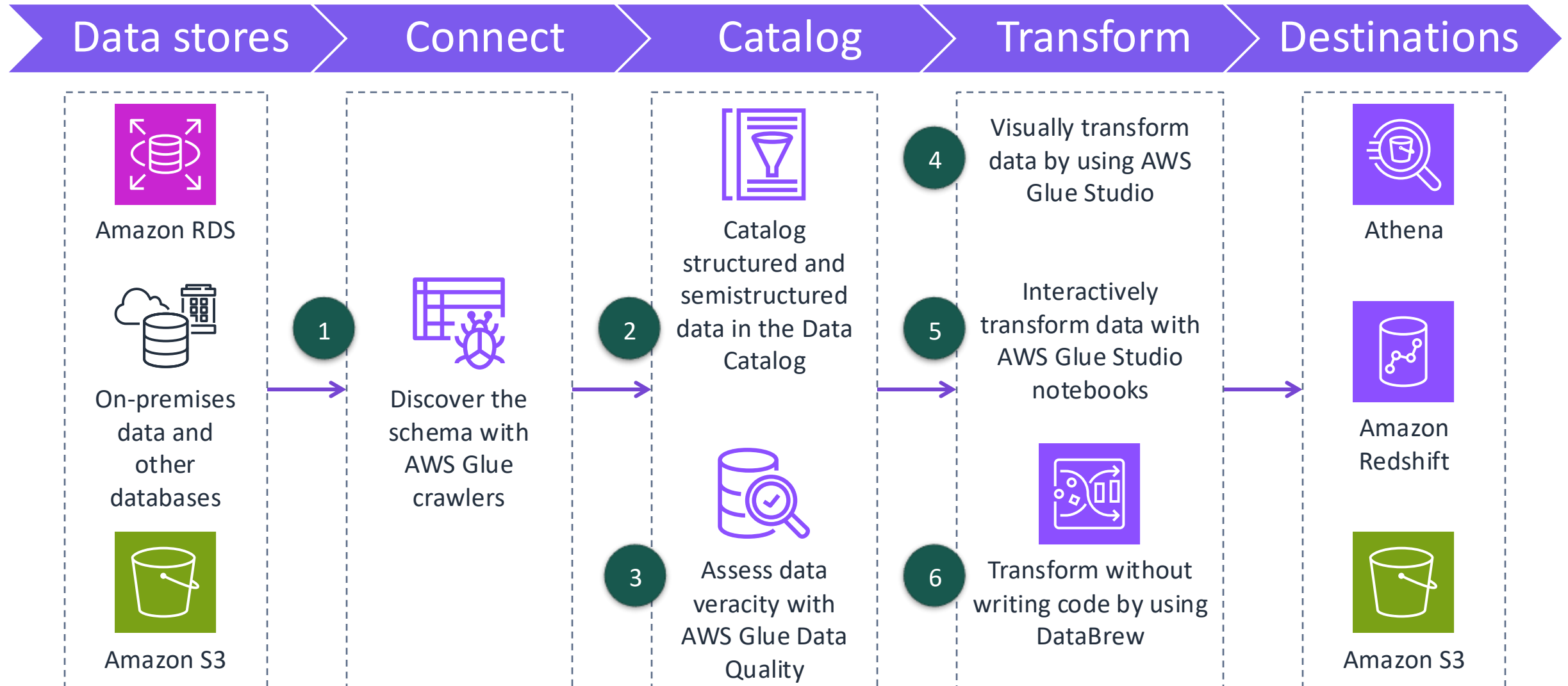
- No-code service to prepare and clean data



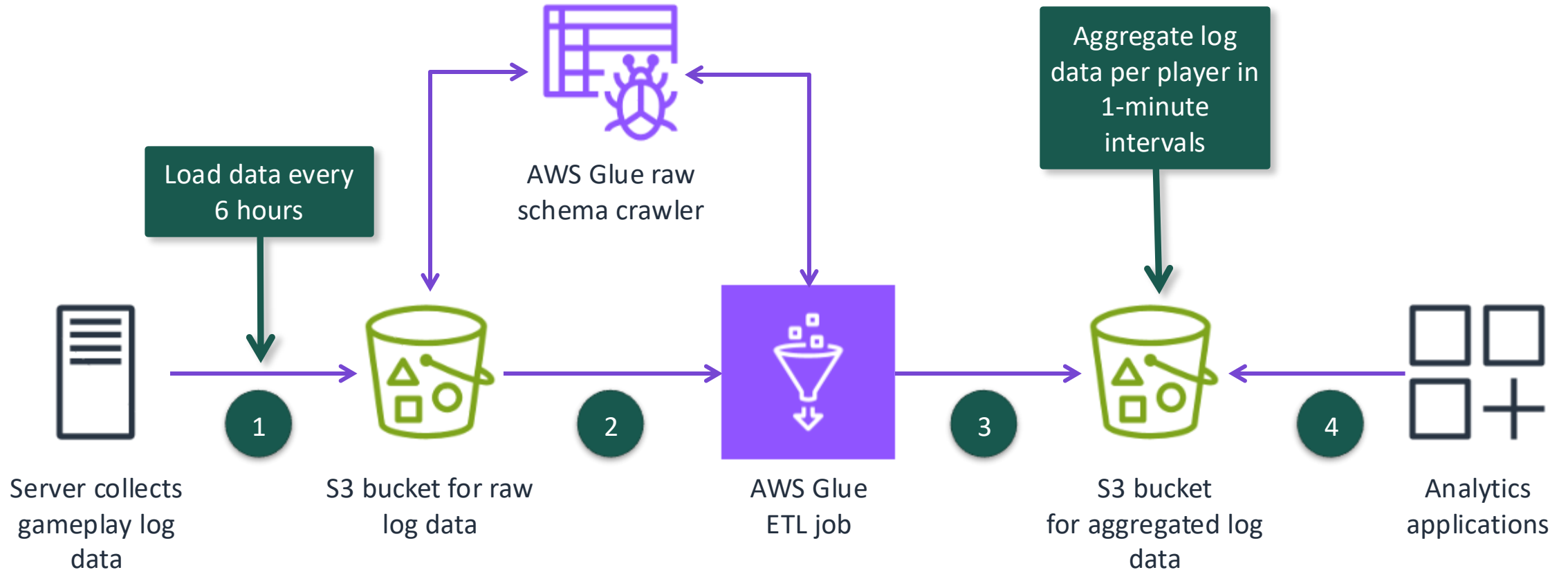
AWS Glue Data Quality

- Data quality assessment

AWS Glue is a data integration service to build modern data architectures



Gaming use case for a batch ETL pipeline



AWS Glue as a data transformation service

.csv

- Is the most common format to store tabular data
- Isn't efficient to store or manipulate large amounts of data (more than 15 GBs)

.parquet

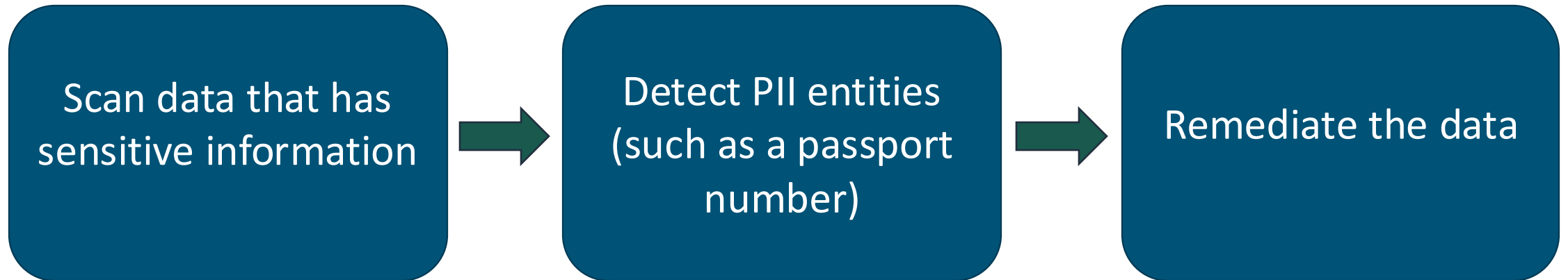
- Stores data in a columnar fashion
- Is optimized for storage
- Is suitable for parallel processing

Convert .csv to .parquet

- Speeds up analytics workloads
- Over time, saves storage space, cost, and time

Advanced transformation using AWS Glue

Transforming data to detect and remove personally identifiable information (PII) before analysis



Key takeaways: Processing batch data



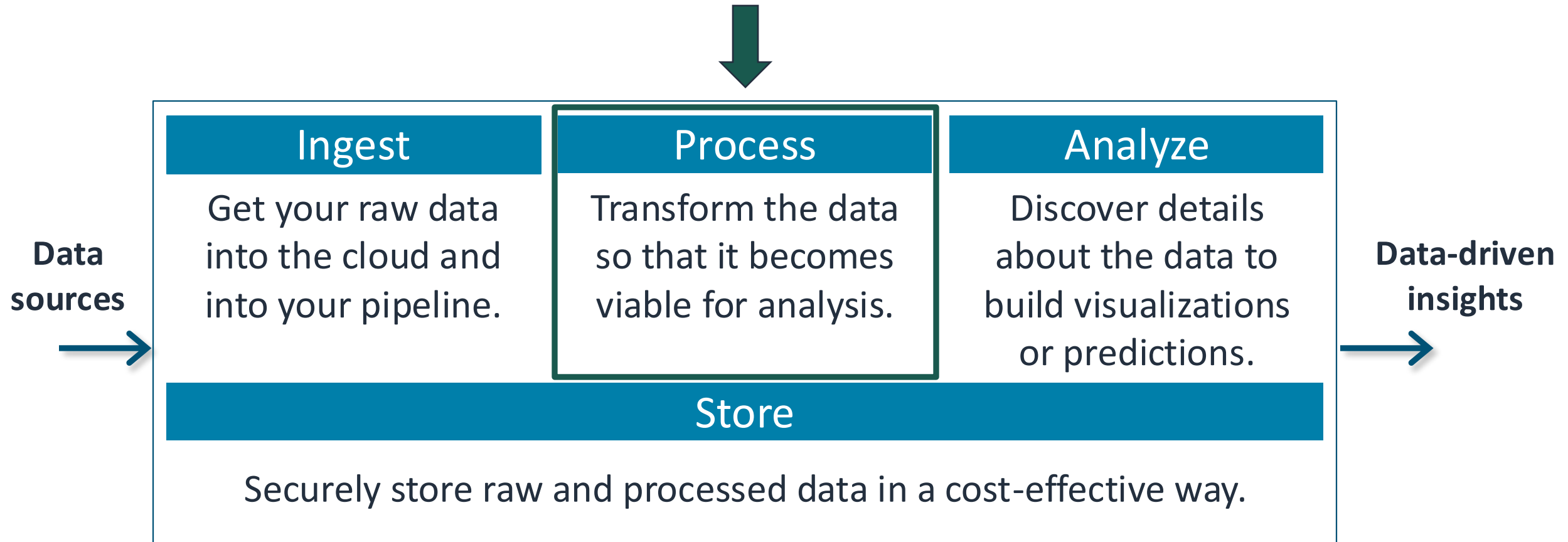
- Batch ingestion and processing make high-volume, repetitive tasks more efficient to run.
- AWS Glue provides functionality for schema identification, data cataloging, data preparation and cleaning, ETL job authoring, and data quality assessment.
- Use AWS Glue when an analytics use case doesn't require real-time aggregation or transformation of data.



Processing real-time data

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Real-time processing layer of the data pipeline



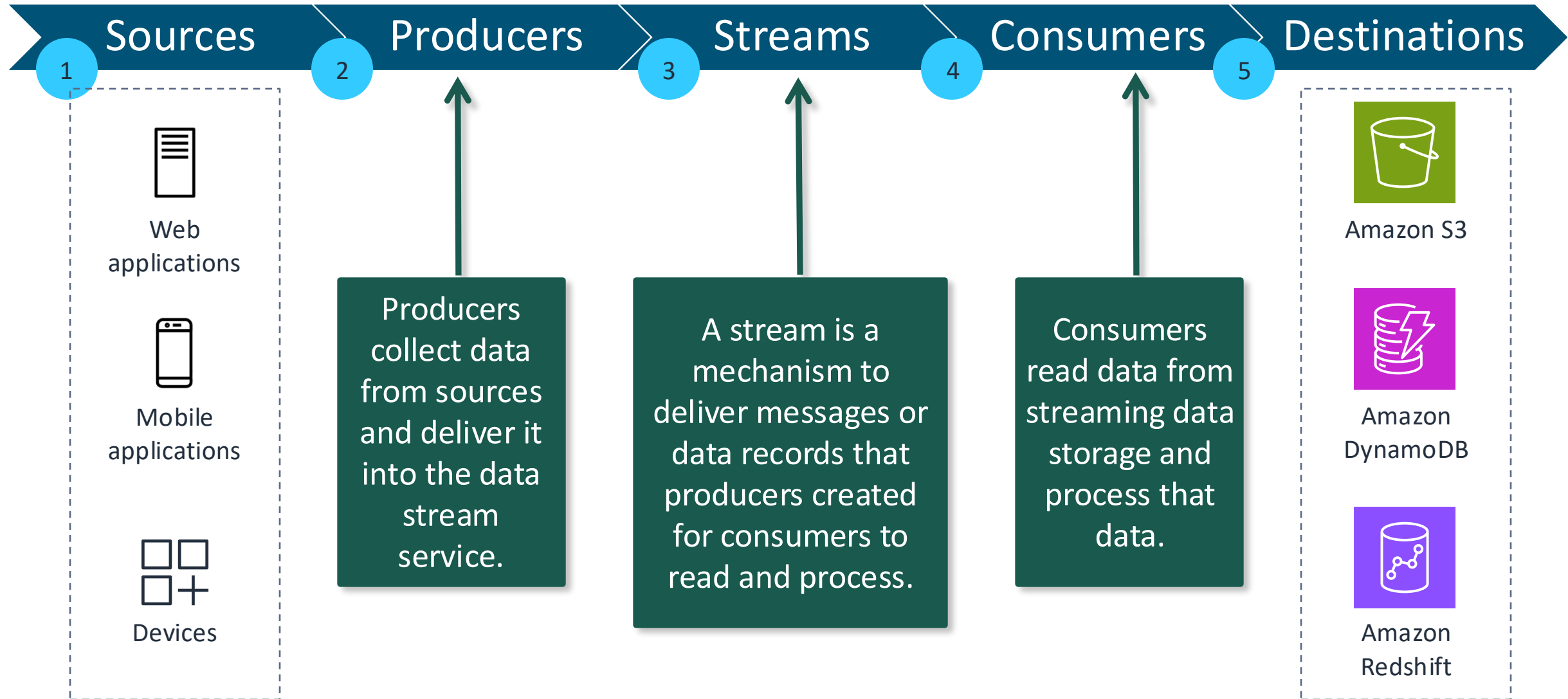
Streaming data

Streaming data is emitted at high volume in a continuous, incremental manner with the goal of low-latency processing.

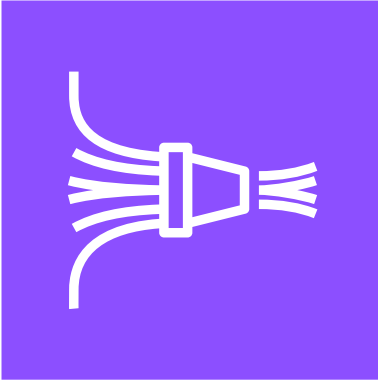
An online gaming company collects streaming data about player-game interactions and feeds the data into its gaming platform. The data gets analyzed in real time and the company offers incentives and dynamic experiences to engage players.



Streaming pipeline workflow and concepts



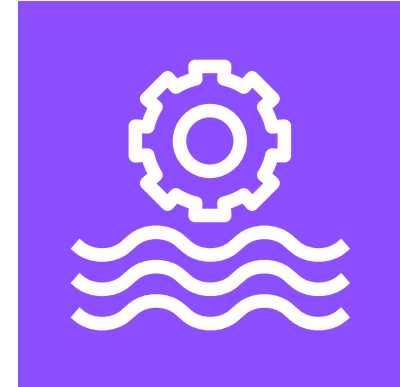
AWS streaming services



Amazon Data Firehose



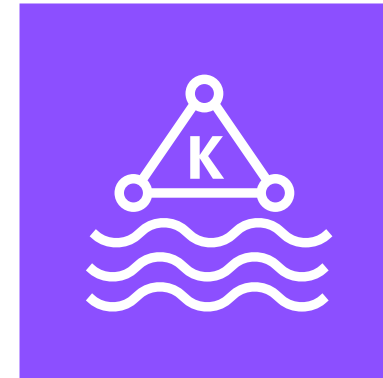
Amazon Kinesis
Data Streams



Amazon Managed Service
for Apache Flink

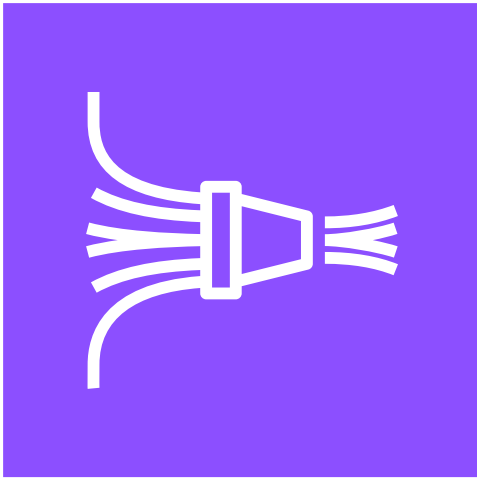


Amazon Kinesis
Video Streams



Amazon Managed
Streaming for Apache
Kafka (Amazon MSK)

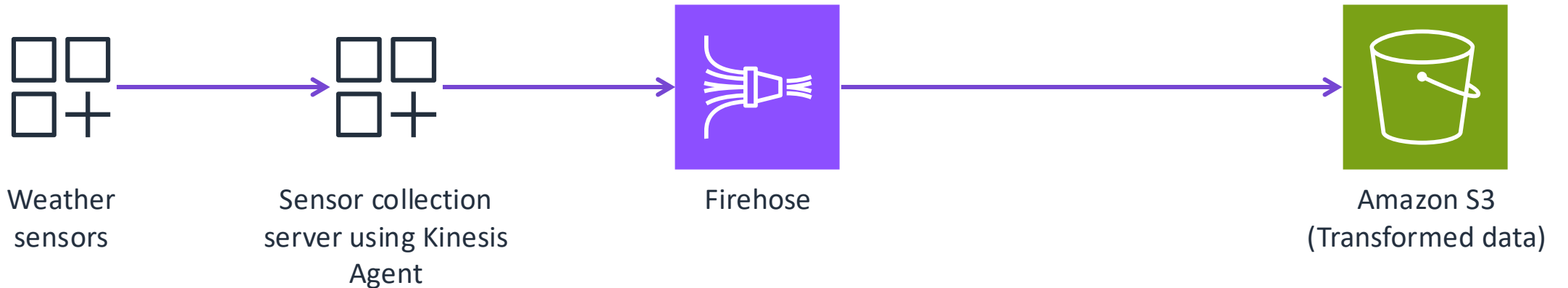
Amazon Data Firehose



Firehose

- Captures and transforms streaming data in near real time
- Is used to deliver data to common storage and analytics destinations
- Delivers data to the storage destination by using micro batch cycles
- Can convert the input data format from JSON to Parquet before storing the data
- Doesn't require coding to use this service

Streaming pipeline workflow using Firehose



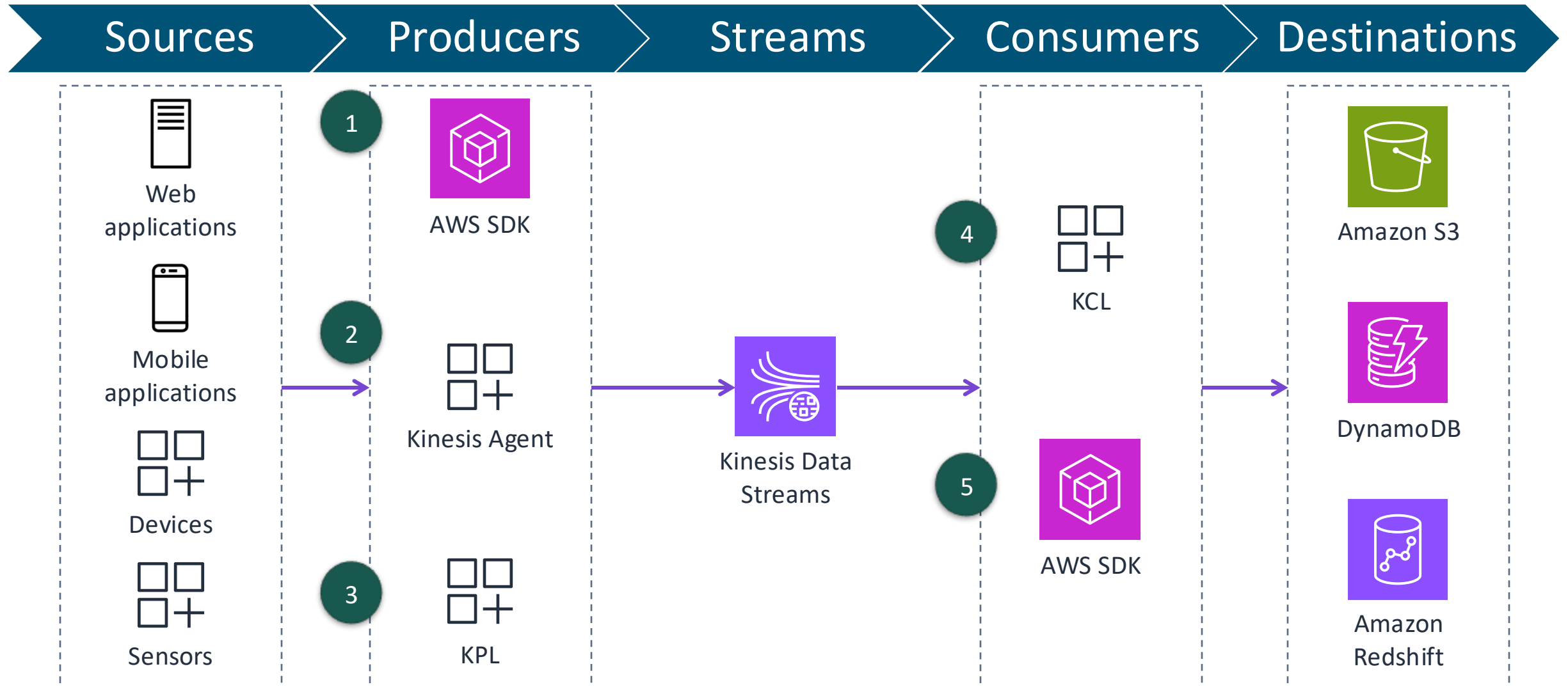
Amazon Kinesis Data Streams



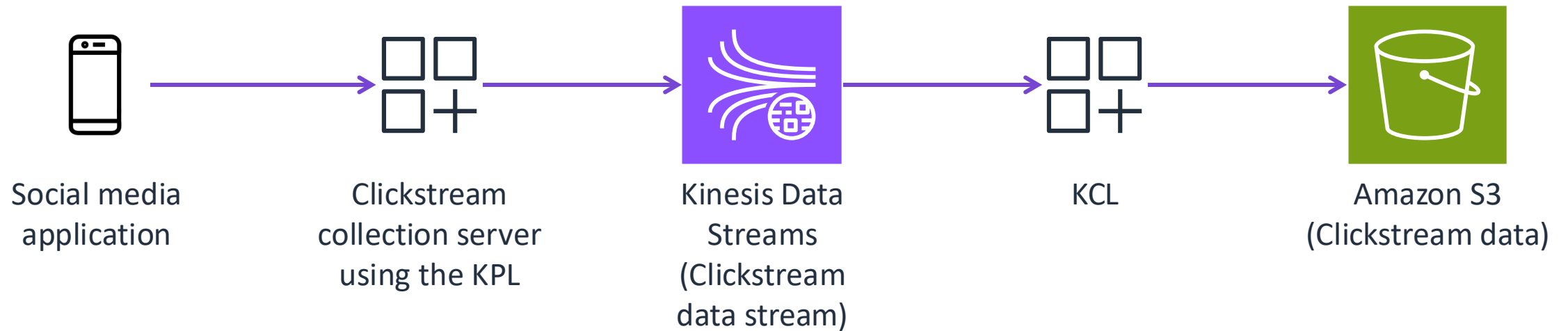
Kinesis Data
Streams

- Ingests stream log and event data
- Provides temporary storage for streaming data (up to 365 days)
- Is used for real-time delivery, within 60 seconds
- Provides the ability for multiple consumers to replay stream data at the same time
- Uses the Kinesis Producer Library (KPL) and Kinesis Consumer Library (KCL) to connect with Kinesis Streams

Producers and consumers in a streaming pipeline



Streaming pipeline workflow using Kinesis Data Streams



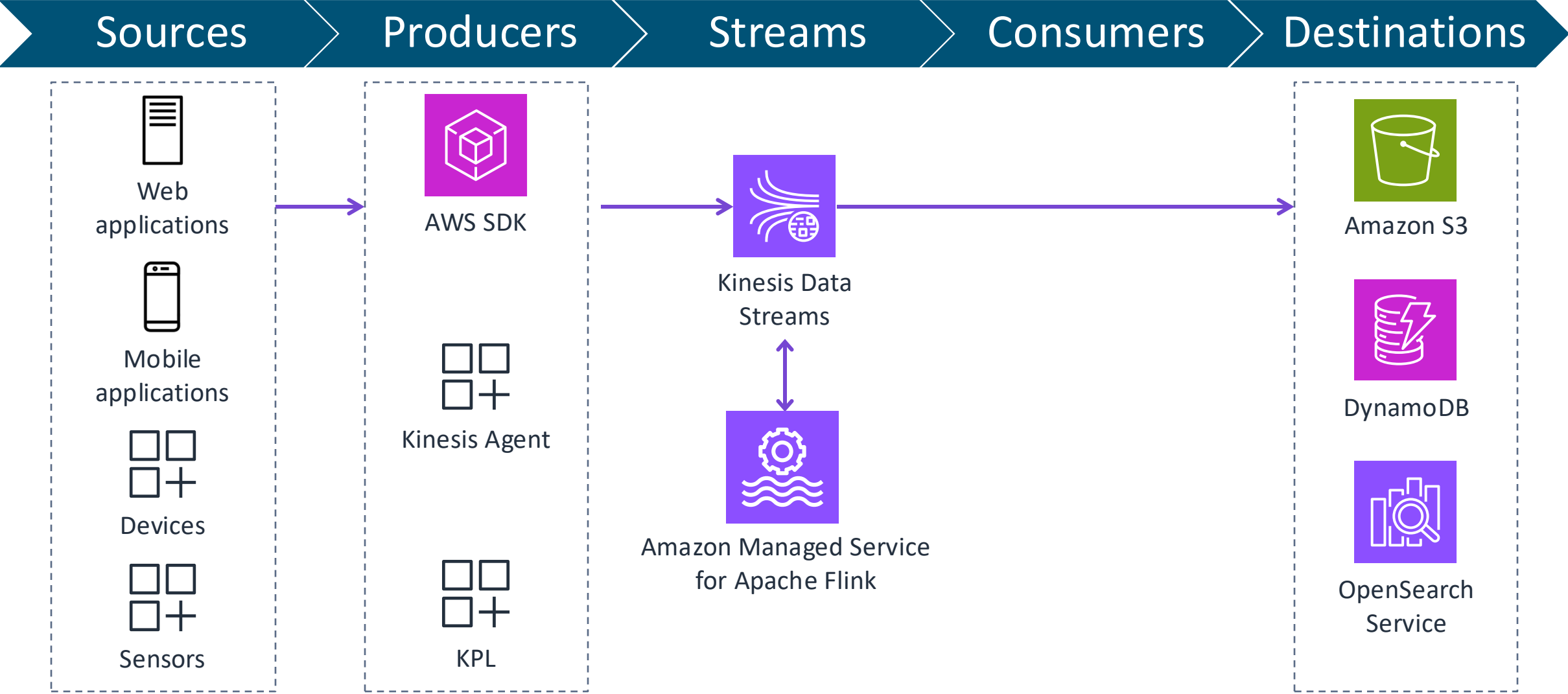
Amazon Managed Service for Apache Flink



Amazon Managed
Service for Apache
Flink

- Provides the ability to build end-to-end processing applications for streaming data
- Processes the data using methods such as aggregation or anomaly detection
- Helps you gain business insights in real time
- Delivers transformed data to a stream or other destination

Processing data with Amazon Managed Service for Apache Flink



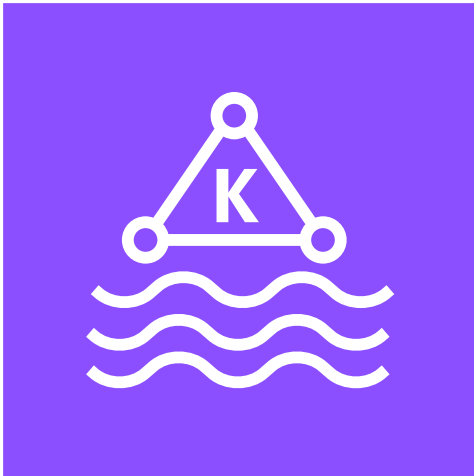
Amazon Kinesis Video Streams



Kinesis Video
Streams

- Securely streams video from connected devices to AWS for analytics, ML, playback, and other processing
- Automatically provisions and elastically scales all the infrastructure that is needed to ingest streaming video data from millions of devices
- Durably stores, encrypts, and indexes video data in your streams and provides access to your data through API operations

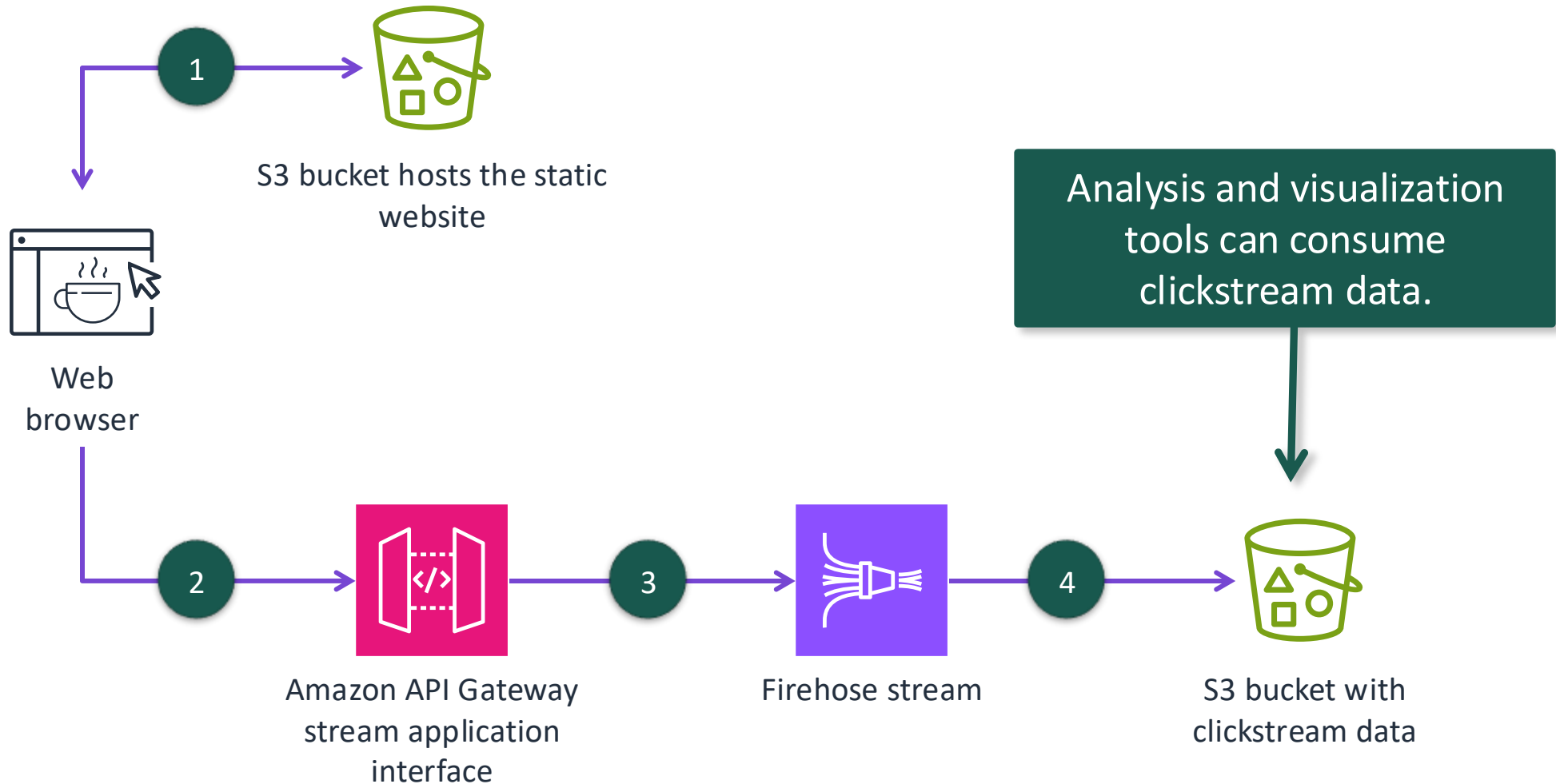
Amazon Managed Streaming for Apache Kafka (Amazon MSK)



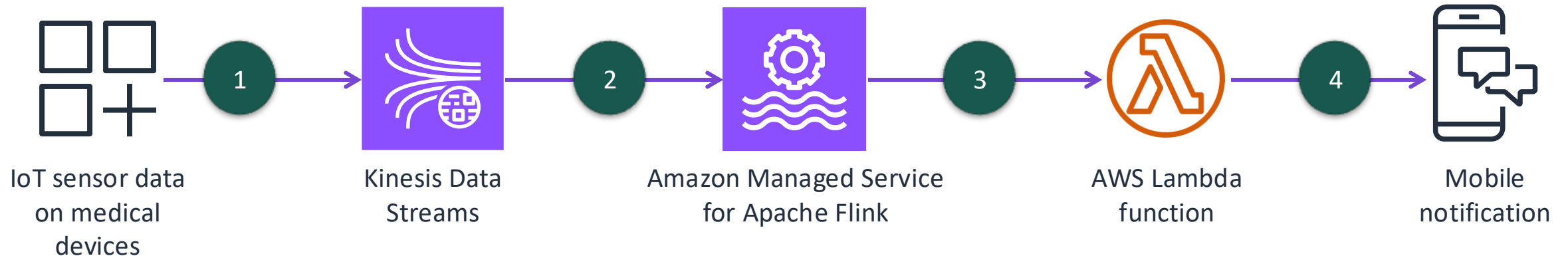
Amazon MSK

- Is a fully managed Apache Kafka service
- Reduces operational overhead
- Can be deployed with existing Apache Kafka tools
- Works with existing AWS integrations

Streaming clickstream data in a café scenario



Medical devices using Kinesis Data Streams and Amazon Managed Service for Apache Flink



Comparison of streaming ingestion services

Firehose

- Is the simplest option to feed data to Firehose storage destinations
- Integrates well with AWS services
- Retains undelivered data for a maximum of 24 hours; doesn't store or replay delivered data

Kinesis Data Streams

- Is an option when you can't use a Firehose approved destination or need more control on how your data is to be consumed
- Requires code customization
- Retains data up to 365 days for replay purposes

Amazon MSK

- Is an open source option to reduce licensing costs; ideal if you're already familiar with Kafka
- Requires you to manage more of the underlying infrastructure
- Retains data longer, and it's configurable

Less complex

More complex

Key takeaways: Processing real-time data



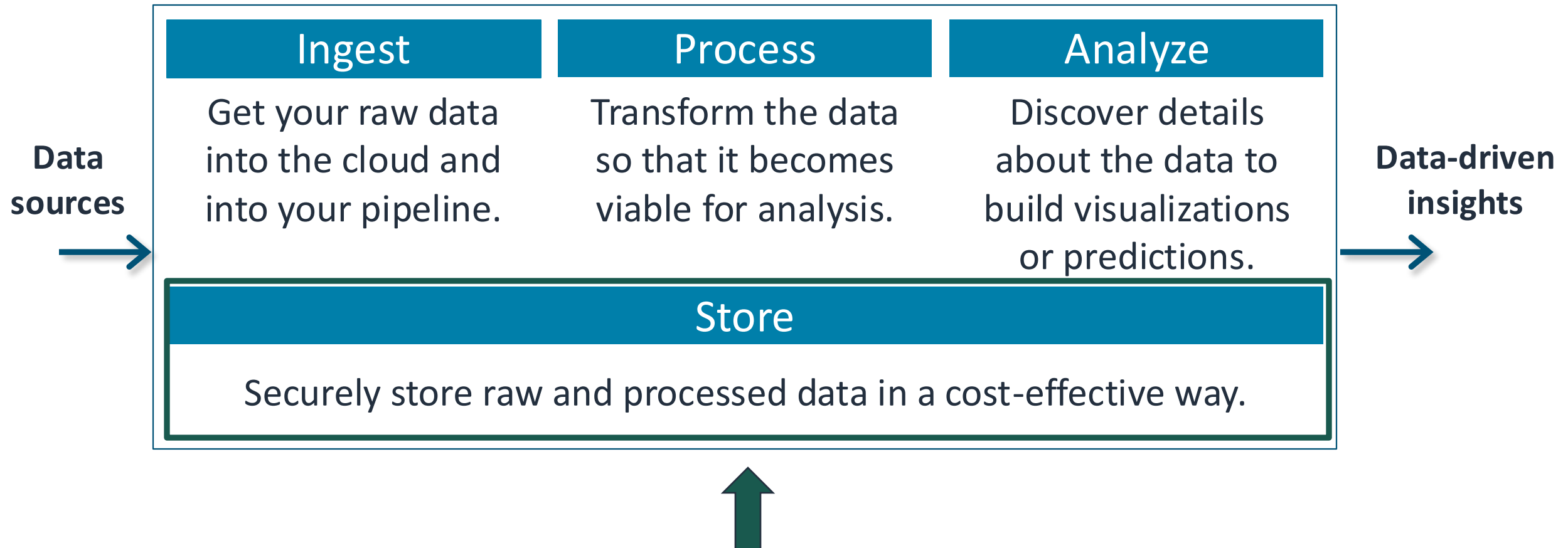
- The Amazon Kinesis family provides fully managed data ingestion services:
 - Firehose delivers near real-time streaming data to certain destinations. It provides short-lived retention of delivered data but lacks replay support.
 - Kinesis Data Streams collects and ingests large streams of data records in near real time. It offers the best low-latency data streaming properties but requires some configuration.
 - Kinesis Video Streams securely streams video from connected devices to AWS for analytics and other processing.
- Amazon Managed Service for Apache Flink analyzes streaming data.
- Amazon MSK is a fully managed Apache Kafka service that you can deploy in a virtual private cloud (VPC).



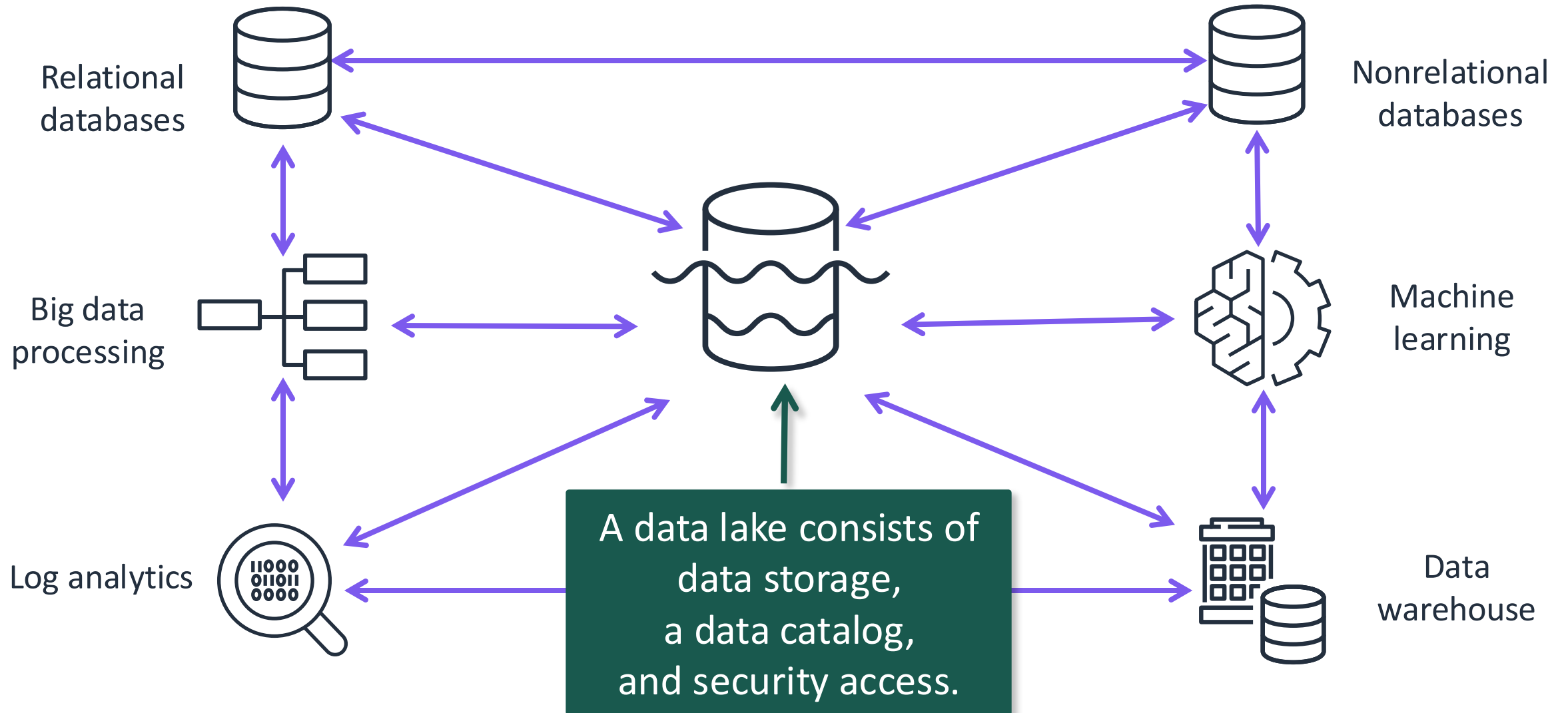
Storage in the data pipeline

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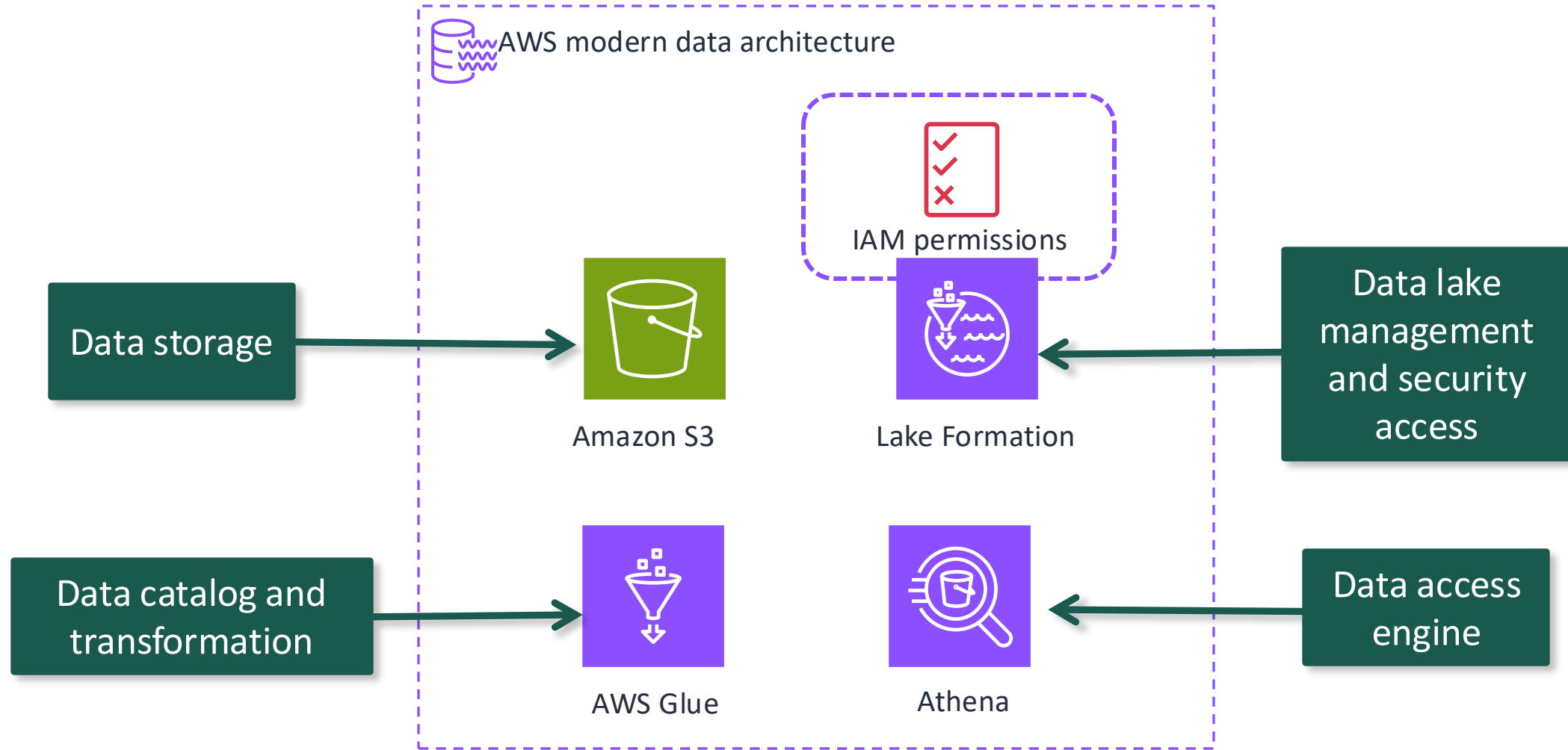
Storage layer in the data pipeline



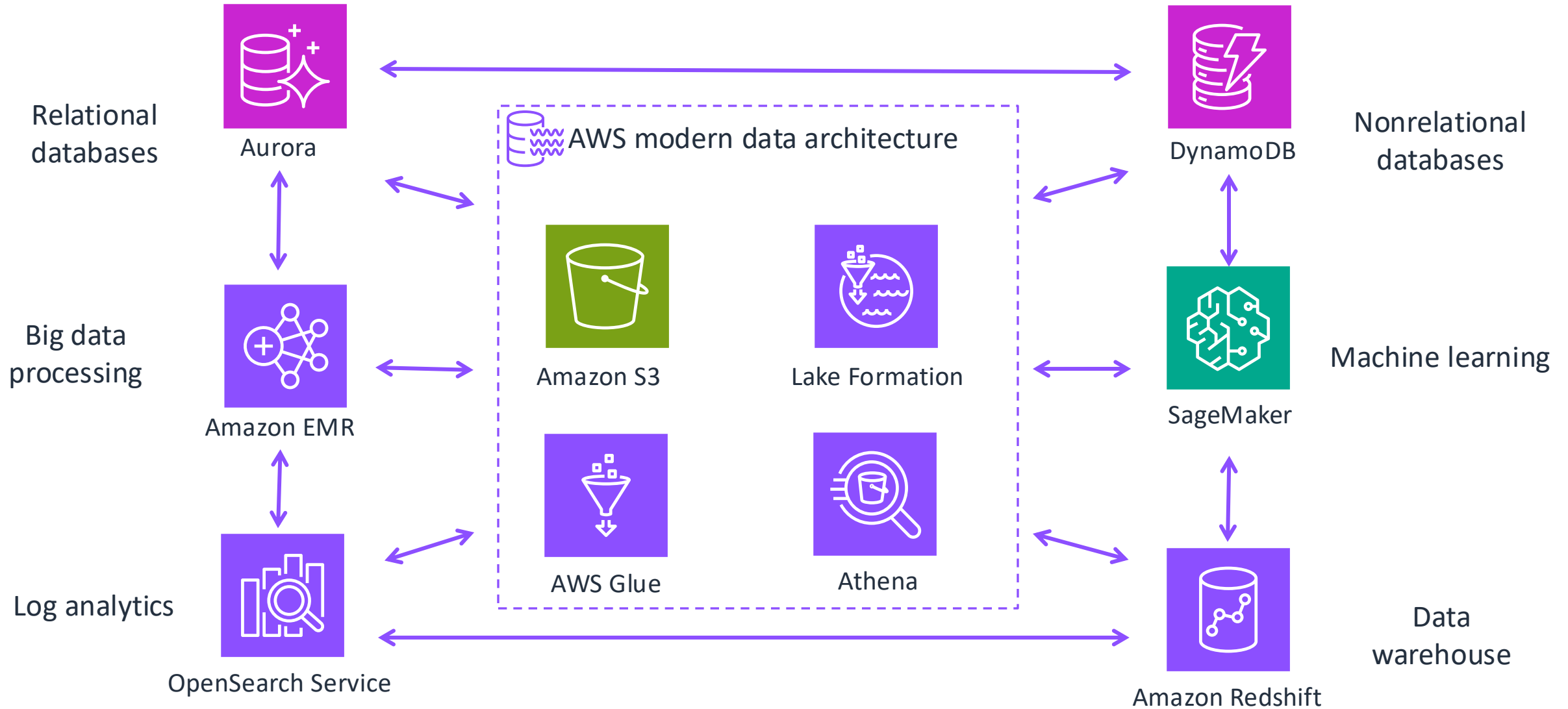
Storage in a modern data architecture



AWS modern data architecture



AWS data lake solution



Activity: Choosing Data Storage for a Bank Application



Banking applications produce the following:

- Customer data
 - Individual customer financial transactions
- Log data
 - User activity logs for fraud analysis
 - Application error logs

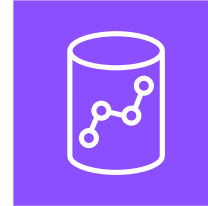
Use case: Storage options for bank application data

Individual financial transactions for customers

Total amount of daily financial transactions for customers

User activity logs for fraud analysis

Error logs for bank applications



Amazon Redshift



Amazon S3



Amazon
DynamoDB

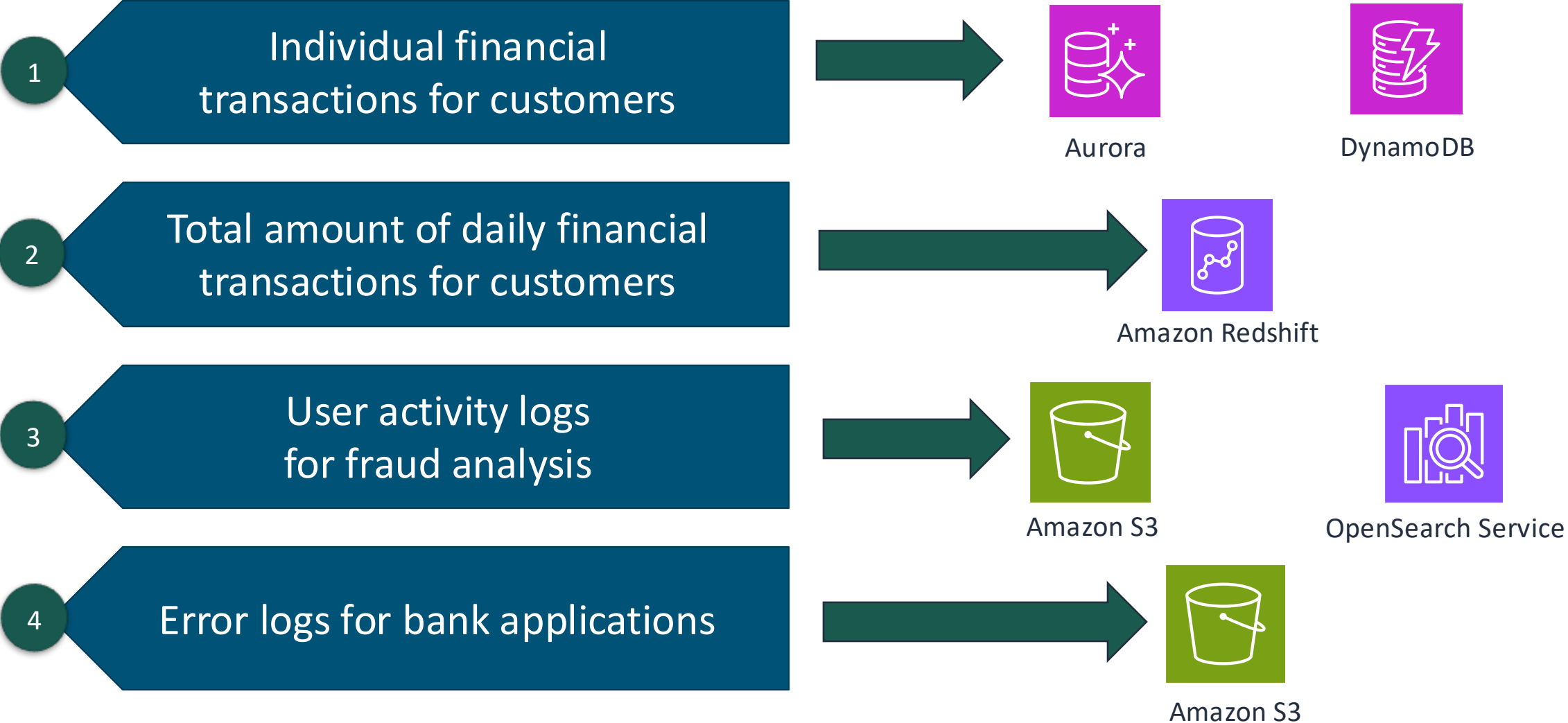


Amazon OpenSearch
Service



Amazon Aurora

Bank application data storage options solution



Comparison of a data warehouse and a data lake

Features	Data Warehouse	Data Lake
Data Sources	Business applications and databases	Internet of Things (IoT) devices, websites, mobile apps, social media, and business applications
Schema	Structured schema on write	Unstructured schema on read
Price	Higher-cost storage	Low-cost storage
Data Quality	Curated, processed, or aggregated data as the central version of truth	Raw (unprocessed data of transactional events) or transformed (processed data)
Analytics Use Case	Batch reporting, business intelligence, and visualizations	Logs, data discovery, and profiling

Cost and querying time determines data storage

Amazon S3

- Lower cost data lake
- Less efficient for querying

Amazon Redshift Spectrum

- Reduces data warehouse cost
- Provides querying on S3 buckets without moving data into the warehouse

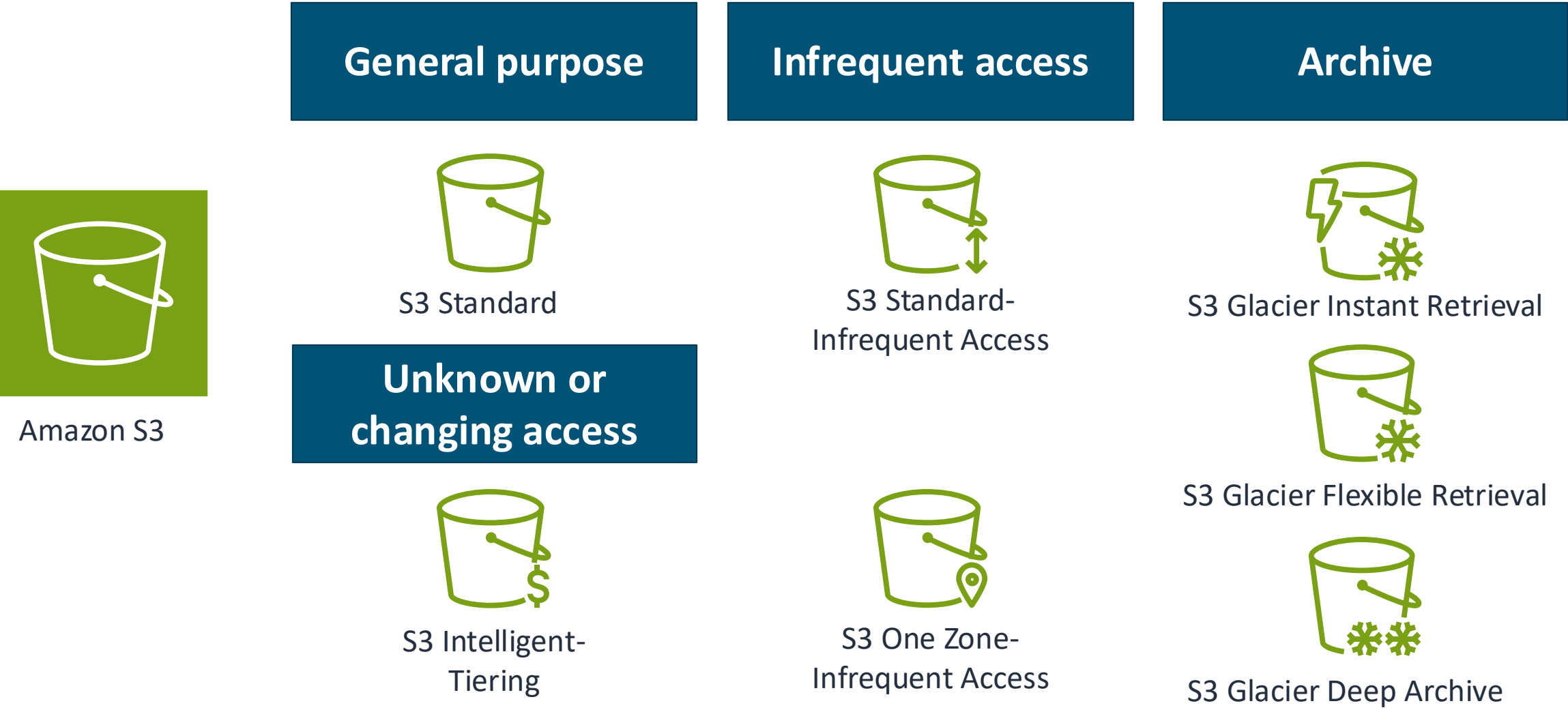
Amazon Redshift

- Higher cost
- Efficient querying across large datasets

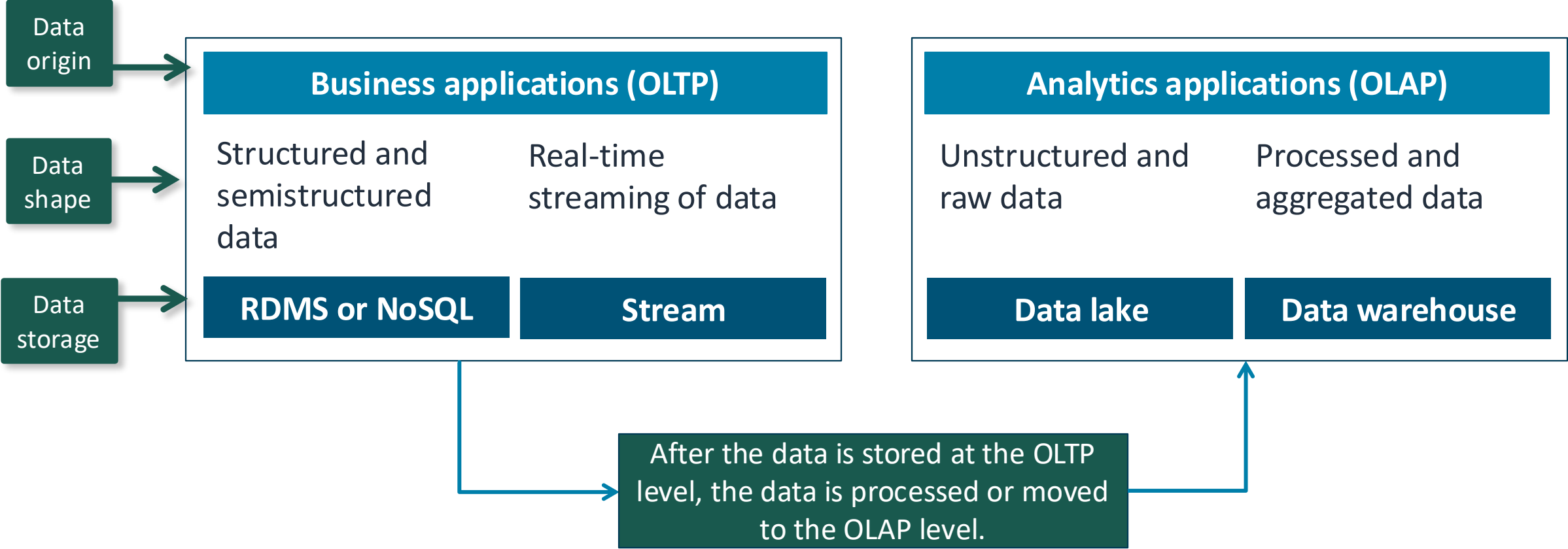


Storage cost

Data retention time determines data storage



Data origin and shape determine data storage



Key takeaways: Storage in the data pipeline



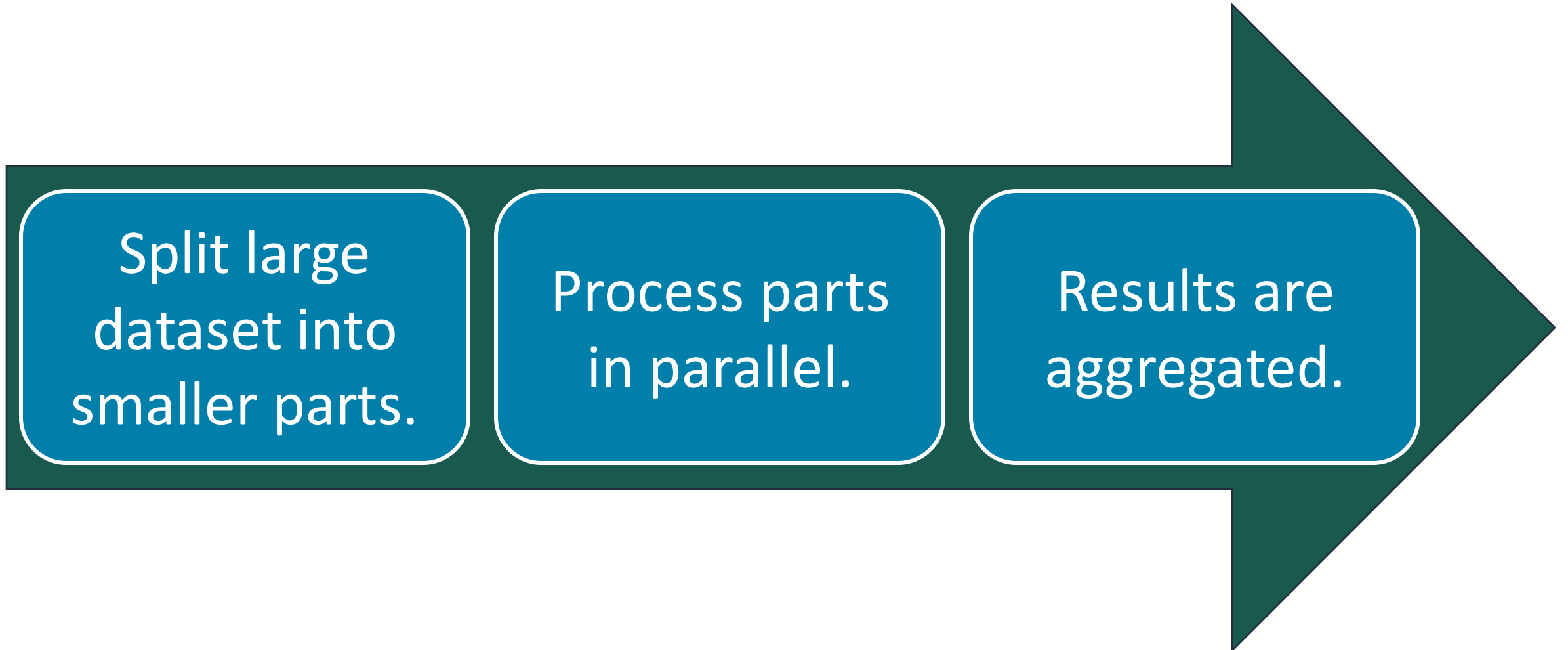
- A modern data architecture is built on a data lake as a central component with analytics and storage services as its peripheral components.
- A data lake architecture consists of data storage, a data catalog, data access security, and data transformation tools.
- A data warehouse has a schema-on-write approach, and a data lake has a schema-on-read approach.
- Design data storage by considering data origin, data shape, cost, query scope, and data retention periods.



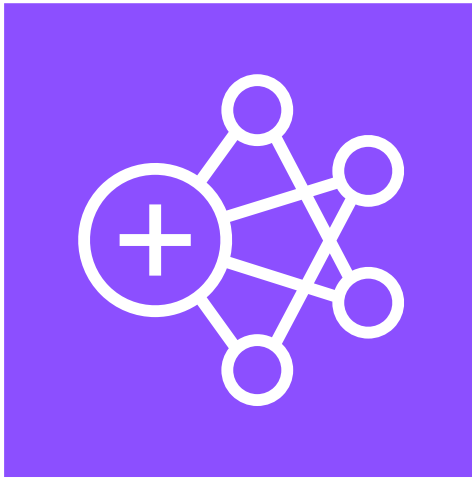
Parallel processing in the data pipeline

Data Engineering Patterns

Big data workflow for parallel processing



Amazon EMR



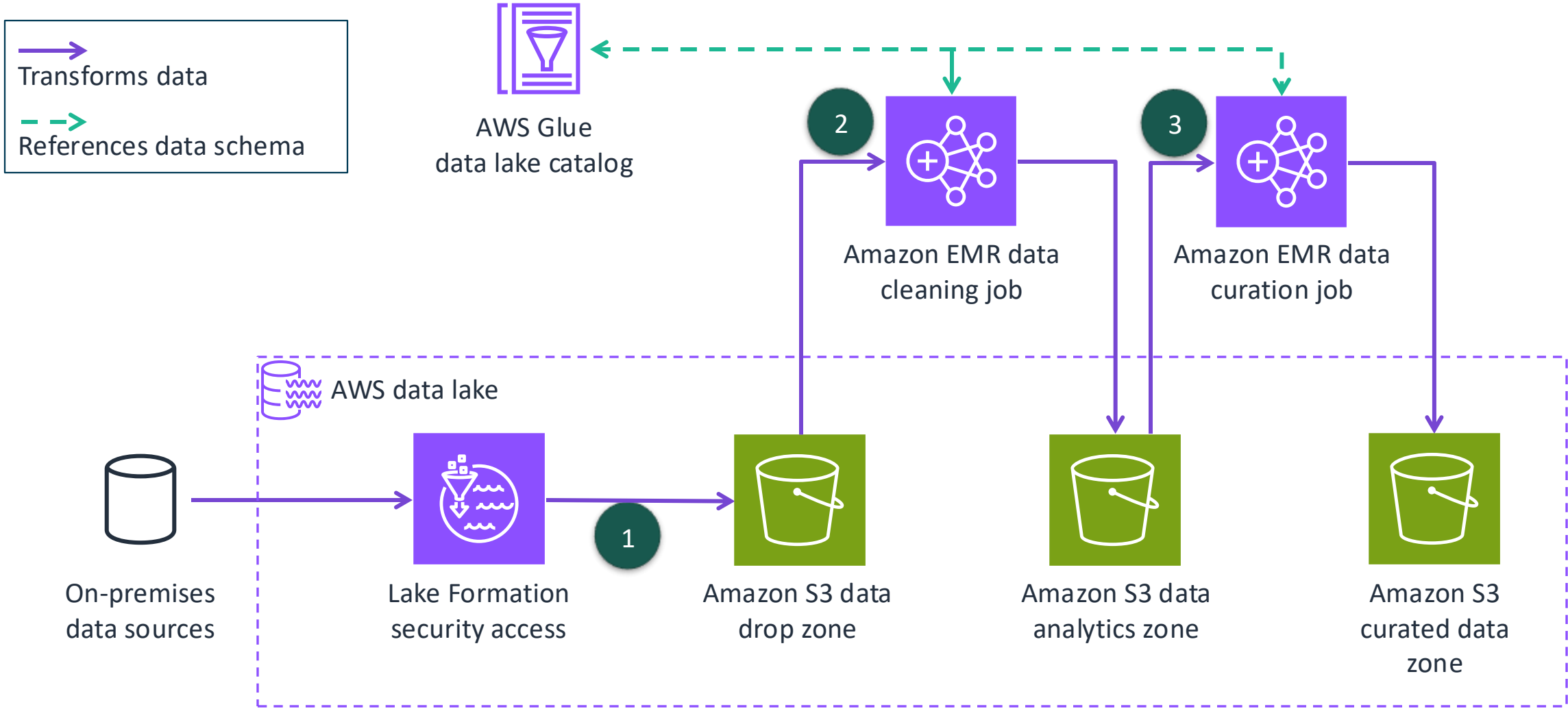
Amazon EMR

- Amazon EMR handles cluster infrastructure management.
- Amazon EMR includes many Apache Hadoop applications, such as Hadoop MapReduce and Apache Spark.
- Amazon EMR can be deployed on Amazon Elastic Compute Cloud (Amazon EC2) instances, Amazon Elastic Kubernetes Service (Amazon EKS), or AWS Outposts.
- Amazon EMR Serverless supports a serverless cluster.

Choosing a parallel processing solution

Requirement	Amazon EMR	EMR Serverless	AWS Glue
Manage and Control Clusters	Yes	No	No
Lift and Shift Legacy Apache Hadoop or Spark Applications to AWS	Yes	No	No
Develop New Cloud Applications for Batch Data Processing	No	Yes	Yes
Has Pay-Per-Job Price Model	No	Yes	Yes
Run Only Apache Spark Jobs	No	No	Yes

Curating data with Amazon EMR



Key takeaways: Parallel processing in the data pipeline



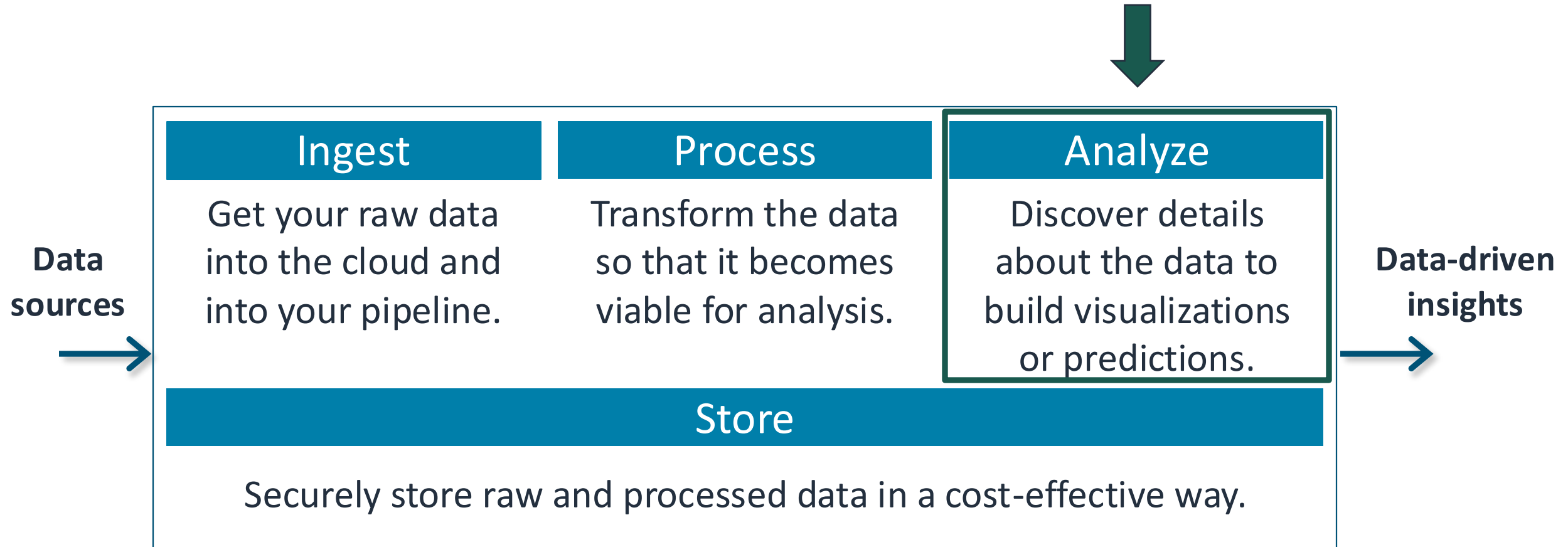
- Big data parallel processing is a computing technique that breaks up the processing of large datasets into smaller parts.
- EMR clusters can use Hadoop MapReduce or Apache Spark frameworks to process data in parallel.
- Choose Amazon EMR to manage clusters.
- Choose EMR Serverless or AWS Glue for serverless solutions.



Analysis and visualization

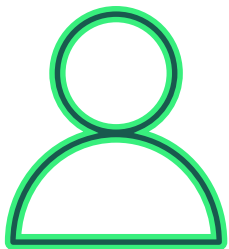
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Analysis and visualization layer of the data pipeline



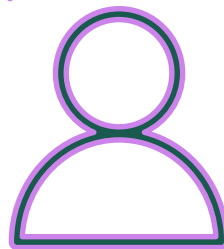
Choosing consumption tools for the data pipeline

I want to design an interactive data dashboard with charts and graphs to share with stakeholders.



Business analyst

I want to use SQL queries to interactively search through customer application activity files in a data lake to solve a one-off customer problem.



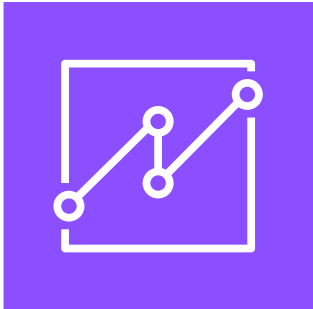
Data scientist

I want to use a real-time application monitoring dashboard.

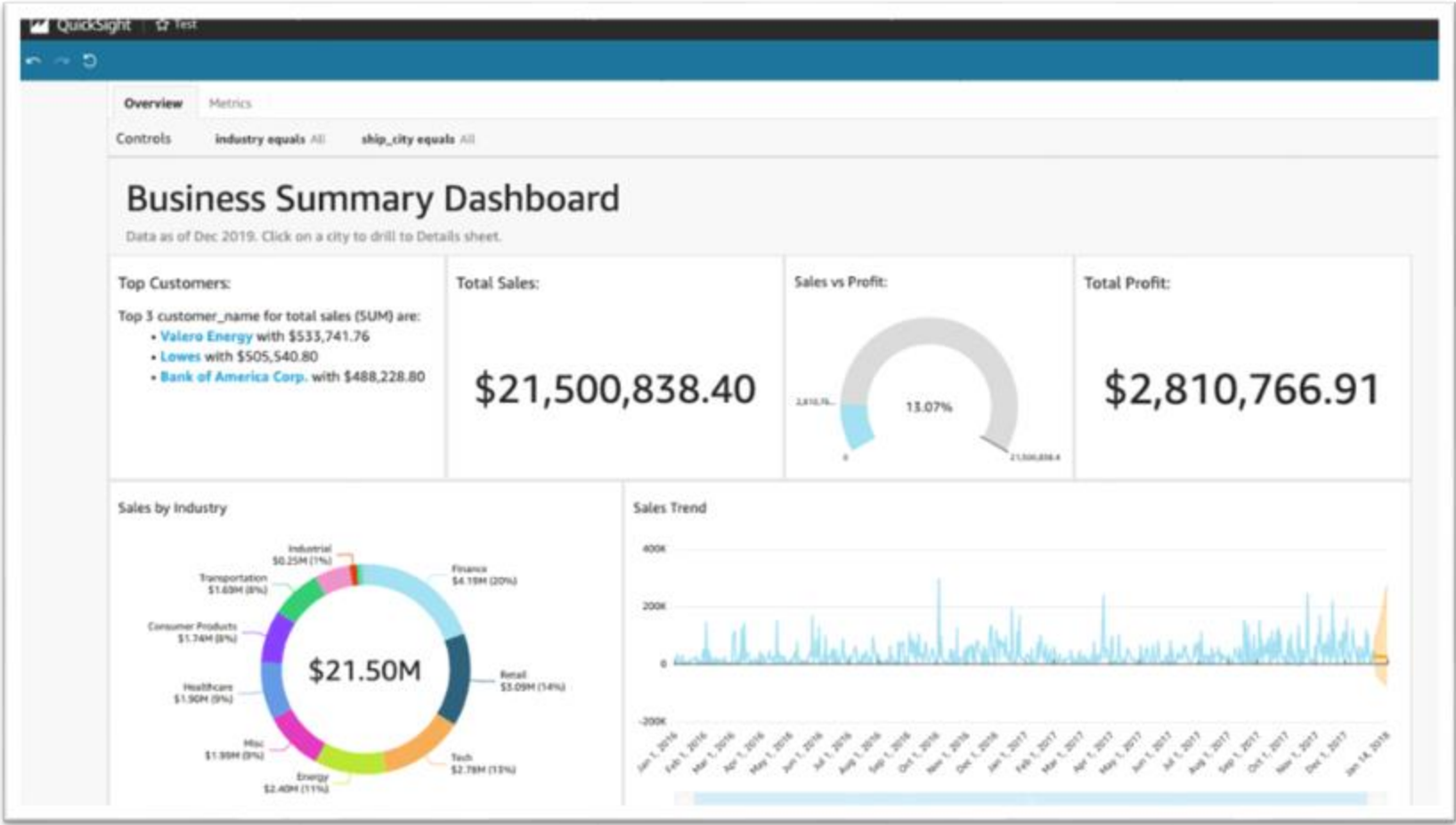


DevOps engineer

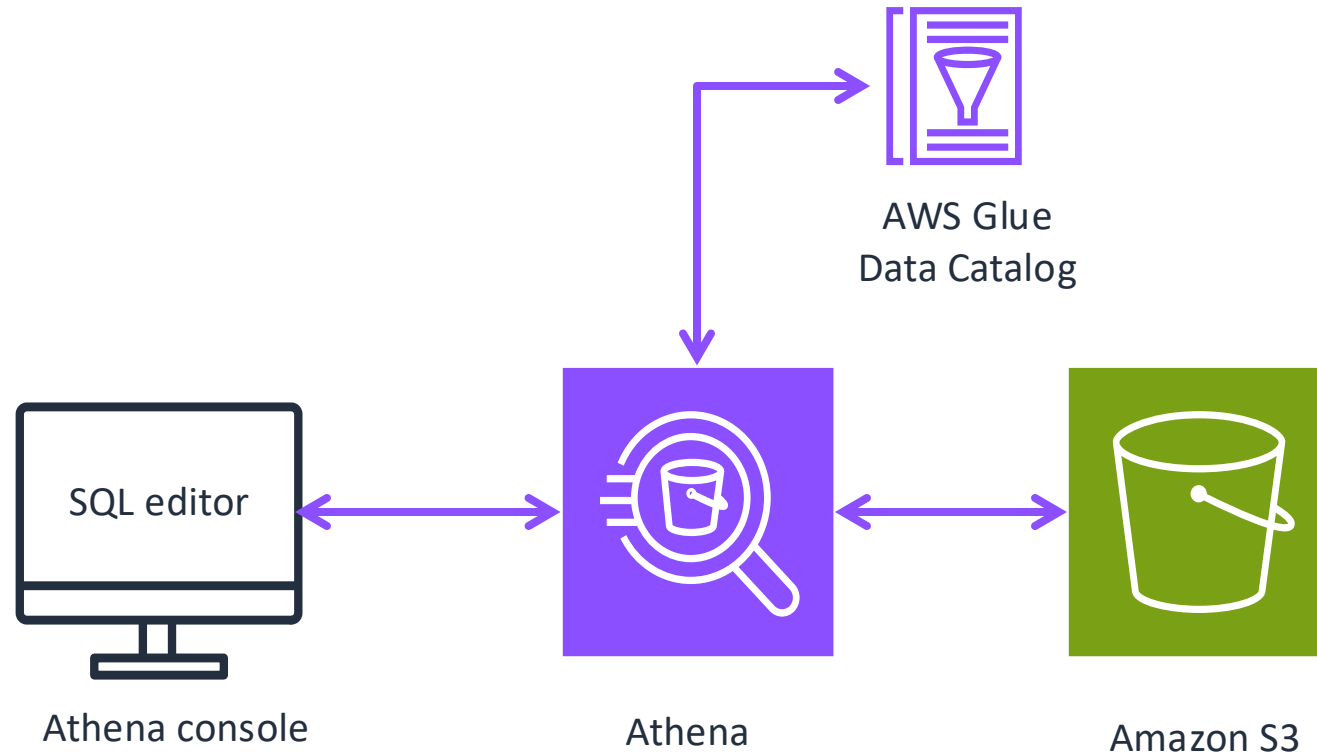
Design dashboards with QuickSight



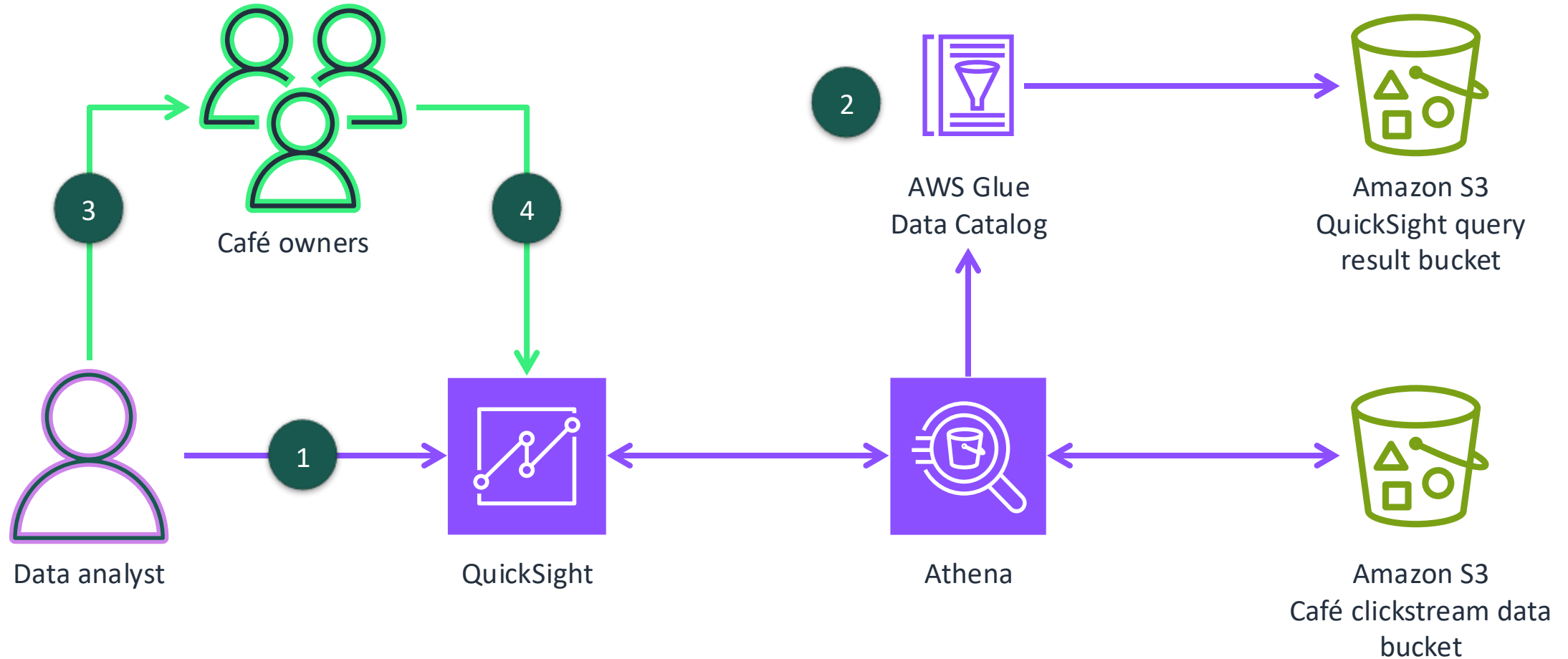
Amazon QuickSight



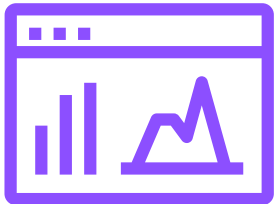
One-time SQL queries with Athena SQL



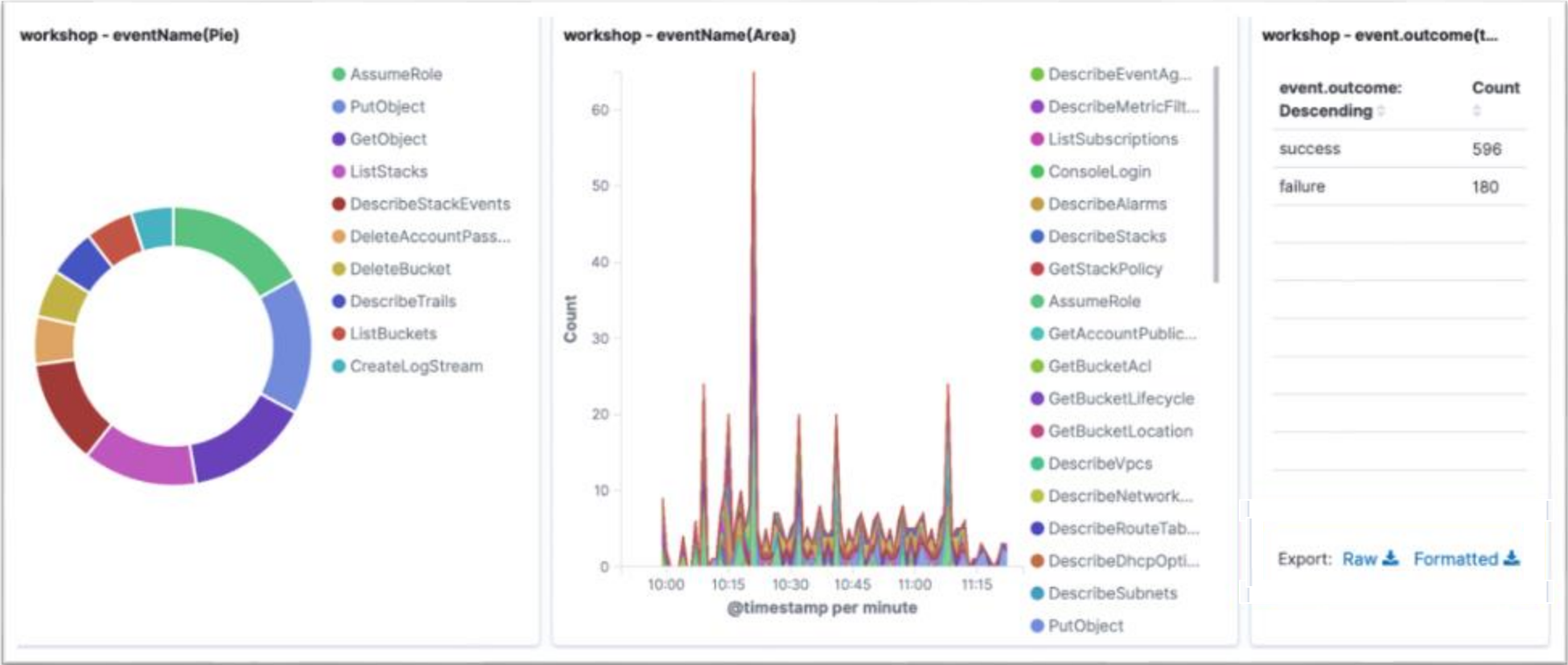
Visualizing clickstream data for the café scenario



Real-time application dashboard with OpenSearch Dashboards



OpenSearch Dashboards



Key takeaways: Analysis and visualization



- QuickSight is a business intelligence tool to visualize data by using publishable dashboards.
- Athena SQL is a data query engine and data schema metadata store with the ability to query data stored in other services by using SQL queries.
- With Athena for Apache Spark, you can use Apache Spark to process parallel workloads of big data.
- OpenSearch Service indexes uploaded data to facilitate search and analysis of data.

Activity: Data Pipeline Architecture



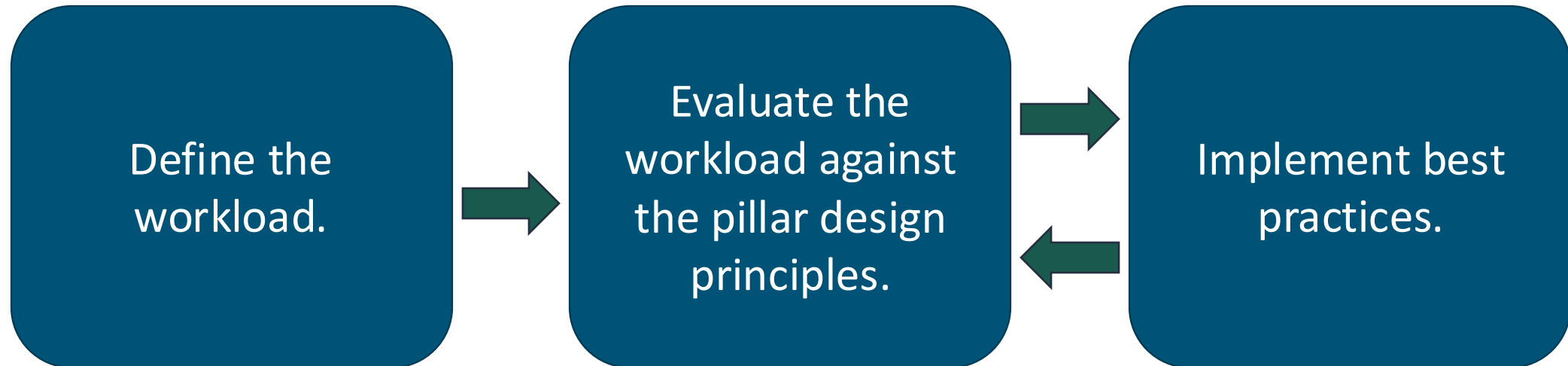
- You are a cloud architect and are assigned to an organization that provides weather data for data analysts.
- Provide an architectural solution for the data pipeline and explain it by using an architecture diagram that you create for a given scenario.



Applying the AWS Well-Architected Framework principles to data pipelines

Data Engineering Patterns

Review process for the Data Analytics Lens



Well-Architected Framework best practices for data analytics



Security



Performance
Efficiency



Cost
Optimization



Reliability

Best practice approach: Control access to the workload infrastructure



Security

Best practice

Implement policies of least privilege for source and downstream systems.

Best practice approach: Choose the best-performing compute solution



Performance
Efficiency

Best practice

Identify analytics solutions that best suit your technical challenges.

Best practice approach: Manage cost over time



Cost
Optimization

Best practices

Remove unused data and infrastructure.

Reduce overprovisioning infrastructure.

Best practice approach: Design resilience for analytics workloads



Reliability

Best practice

Understand the business requirements of analytics and ETL jobs.

Key takeaways: Applying the AWS Well-Architected Framework to data pipelines



- Implementing the Data Analytics Lens is a continual process with regular evaluations to add as many best practices as possible to an analytics workload.
- Prioritize the pillars in order of importance, and identify the most important design principles to implement.



Module wrap-up

Data Engineering Patterns

Module summary

This module prepared you to do the following:

- Use the Well-Architected Framework to generalize the type of architecture that is required to suit common use cases for data ingestion (batch and stream).
- Select a data ingestion pattern that is appropriate to the characteristics of the data (velocity, volume, and variety).
- Select the appropriate AWS services to ingest and store data for a given use case.
- Select the appropriate AWS services to optimize data processing and transformation requirements for a given use case.
- Identify when to use different types of AWS data analytics and visualization services based on a given use case.

Considerations for the café

Discuss how you, as a cloud architect, might advise the café based on the key concerns for cloud architecting presented at the start of this module.



Module knowledge check



- The knowledge check is delivered online within your course.
- The knowledge check includes 10 questions based on material presented on the slides and in the slide notes.
- You can retake the knowledge check as many times as you like.

Sample exam question

A data engineer needs to analyze and visualize a lot of streaming data from user activity logs in real time. Which single service would provide this capability?

Identify the key words and phrases before continuing.

The following are the key words and phrases:

- Analyze and visualize
- Streaming data logs
- Real time
- Single service

Sample exam question: Response choices

A data engineer needs to **analyze and visualize** a lot of **streaming data** from **user activity logs** in real time. Which single service would provide this capability?

Choice	Response
A	Amazon OpenSearch Service
B	Amazon Athena
C	Amazon QuickSight
D	Amazon Redshift

Sample exam question: Answer

The answer is A.

Choice	Response
A	Amazon OpenSearch Service



Thank you

Corrections, feedback, or other questions?

Contact us at <https://support.aws.amazon.com/#/contacts/aws-academy>.



Appendix

Data Engineering Patterns

Appendix: Data characteristics

- Value of data
- Veracity of data
- Volume of data
- Velocity of data
- Variety of data

Appendix: Value of data

Scope	Ask	Consider
• Value is about how an organization can use processed data to gain insight to a business problem or solve it.	• What insights can you gain from the data? • How should you process the data to get the answers that you need?	• Keep the end user in mind as you design the infrastructure and make decisions.

Appendix: Veracity of data

Scope	Ask	Consider
• Veracity is about understanding the full lifecycle of your data and knowing how to protect and strengthen the integrity of your data.	• How accurate, precise, and trusted is the data?	• Value rests on veracity because without good data, you could make bad business decisions.

Appendix: Volume of data

Scope	Ask	Consider
• Volume is about how much data you need to process.	• How big is the dataset? • How much new data is generated? • How long do you need to keep the data? • How often do you need to access the data?	• Understand the duration of keeping the data and the access frequency to weigh the cost and benefits of storage.

Appendix: Velocity of data

Scope	Ask	Consider
• Velocity is about how quickly data enters and moves through your pipeline.	• How frequently is data generated? • How quickly does the data need to be acted upon? • Can your pipeline handle sudden bursts of data?	• Volume and velocity together drive the expected throughput and scaling requirements of your pipeline.

Appendix: Variety of data

Scope	Ask	Consider
• Variety is about the different types of data and the number of diverse data sources you deal with.	• What is the format of the data? • What is the data type? Is it structured, semistructured, or unstructured? • How many different sources does the data come from?	• Combining datasets can enrich analysis but can also complicate processing.

Appendix: Data pipelines

- Transforming data during ingestion

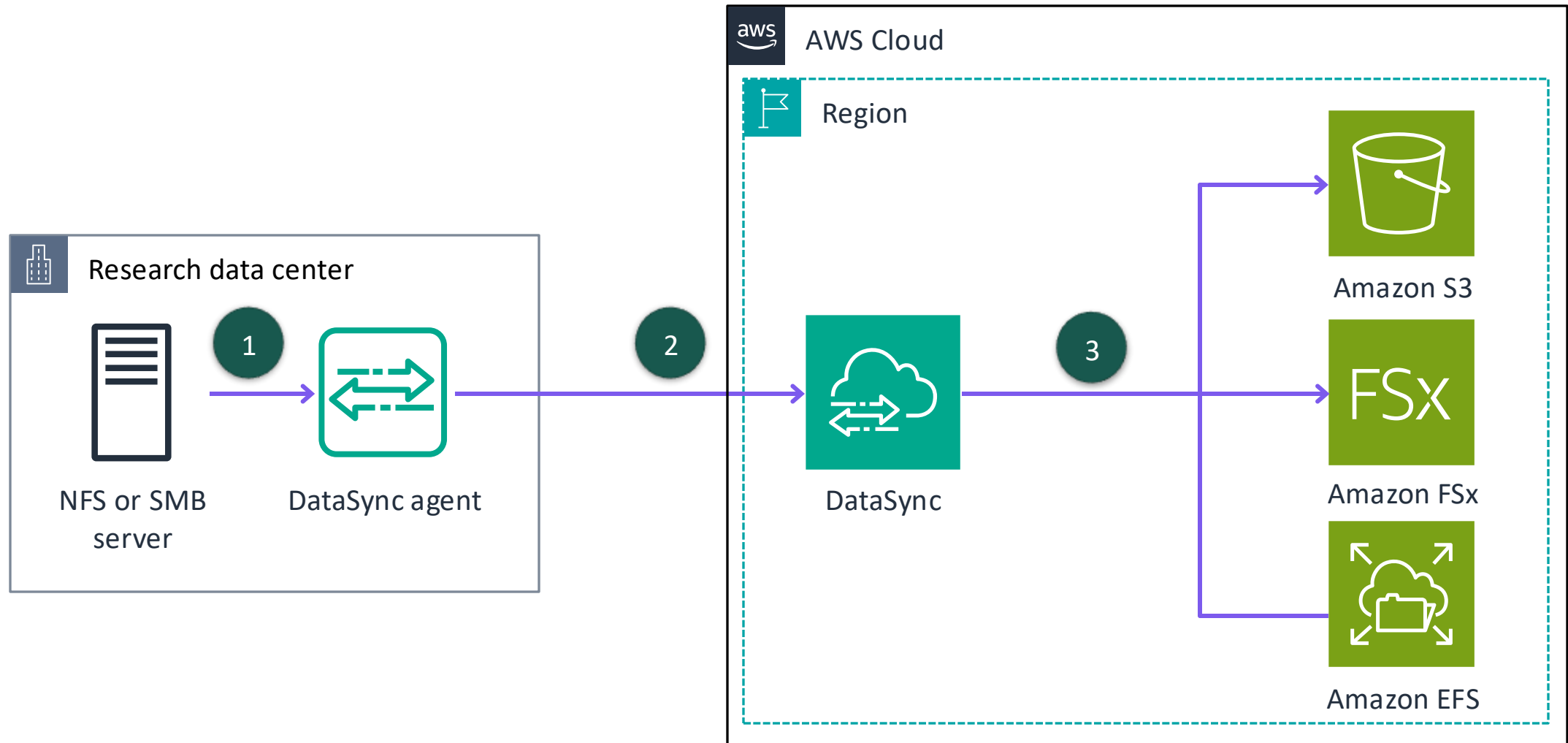
Appendix: Transforming data during ingestion

- Raw data can be inconsistent, imprecise, and repetitive. Data will almost always be transformed as it moves through the pipeline.
- You might need to prepare the data before it's viable for analysis:
 - Clean or normalize the data.
 - Modify the format to support a specific analysis tool.
 - Augment the dataset by filling in gaps or enrich it with additional information.
 - Add metadata to categorize and catalog data.

Appendix: AWS tools to ingest data

- AWS DataSync: Transfer between on premises and AWS

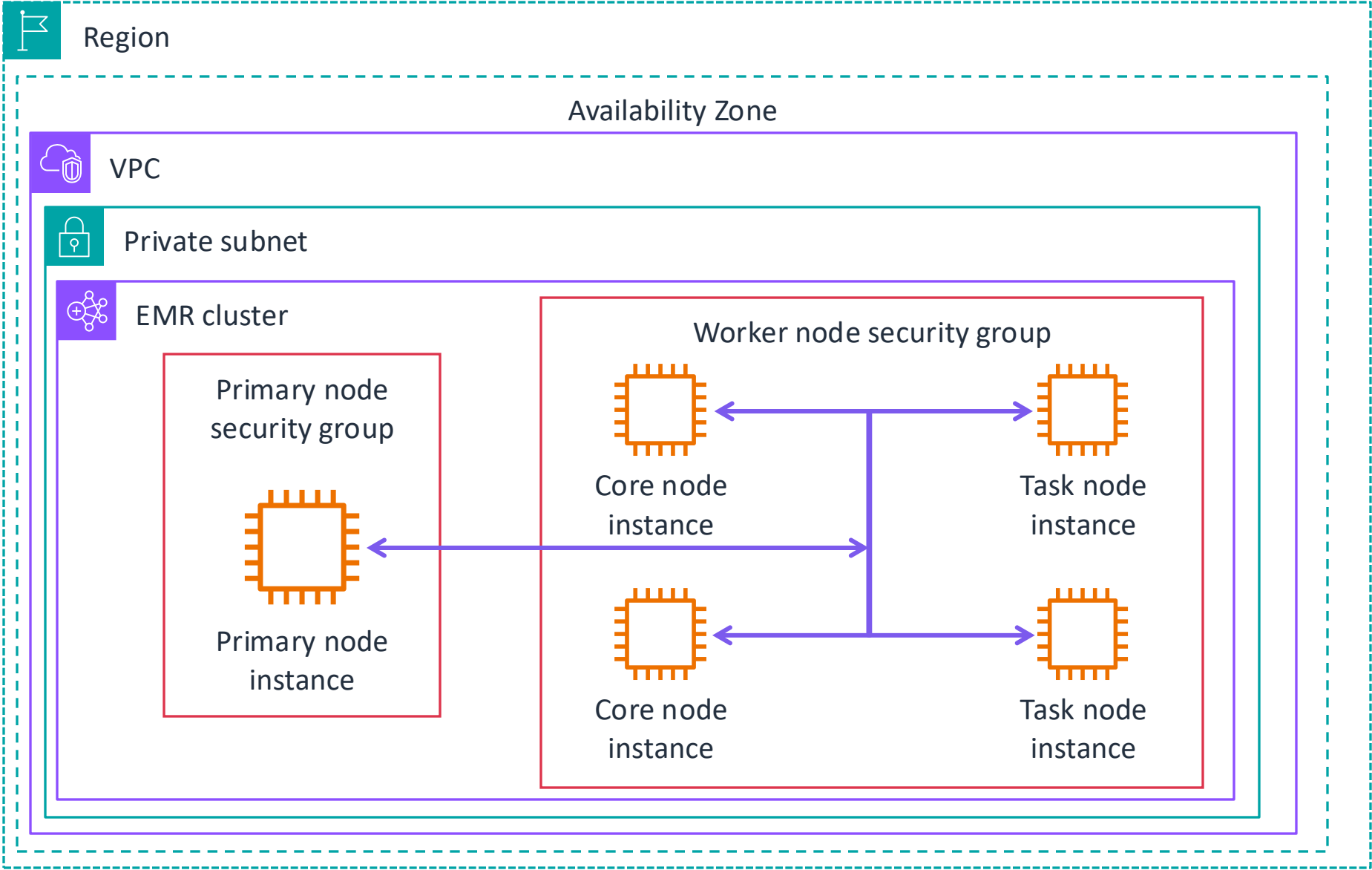
AWS DataSync: Transfer between on premises and AWS



Appendix: Processing real-time data

- EMR cluster architecture on Amazon EC2
- Amazon EMR workflow

EMR cluster architecture on Amazon EC2



Amazon EMR workflow

